

# **A Case Study of Analyzing Wind Data Proposing a Wind Farm Design to Provide Zero-Carbon Energy to Hettinger City.**

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## Abstract

Analyzing wind data of Hettinger City, North Dakota, to design a wind farm that can utilize renewable energy to generate electricity, thus increasing the sustainability of the area. This report accounts for environmental impact, licenses, zoning regulations, installation costs and capacity factor in the process of the design, thus providing the complete outline involving the design process.

## Introduction

The area chosen for the analysis and design of our wind farm was Hettinger City in North Dakota seen in Figure 1. According to the United States Census Bureau, the population is roughly 1,051 as of July 2021. The state of North Dakota ranks top 10 in average wind speeds which makes it a great place for wind farms that can power entire cities and even potentially provide extra power back to the grid. This report shows the production for 100% of the electricity load of Hettinger which is roughly 32 GWh.



Figure 1: Hettinger City, ND

## Electrical Energy Demand

According to Find Energy, the total energy consumption in Hettinger was 32.830 GWh as of 2021. With a population of 1050, this consumption is split between industrial, commercial and residential. Demand does not remain constant through the year due to seasonal changes in electricity demand. Seen below in Figure 2 is a monthly energy estimation of Hettinger. As seen in the chart, energy consumption is higher during the winter due to increased heating demands.

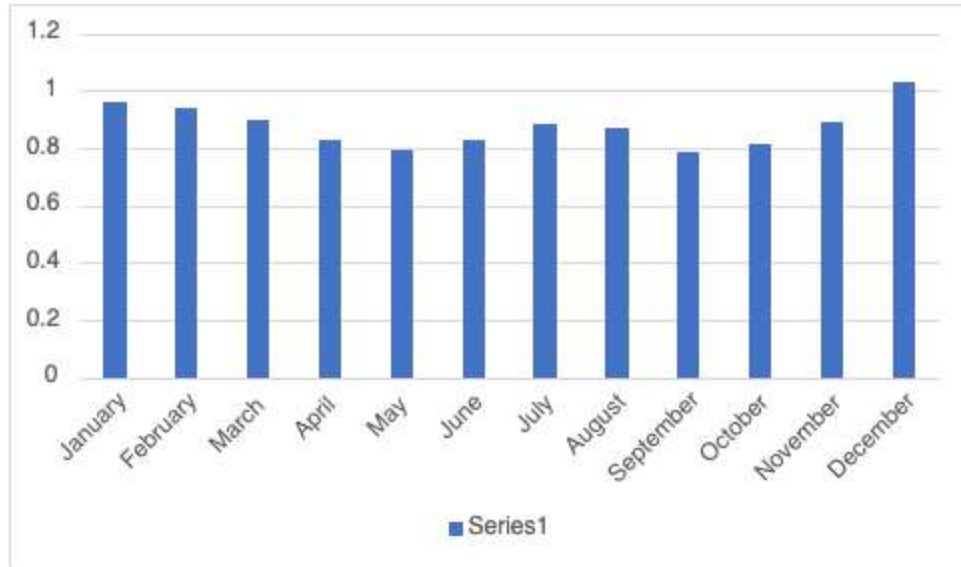


Figure 2: Estimated Monthly Energy Demand

### Wind Data Analysis

The raw 50M and shear exponent calculations for 100M are provided in the reference to our raw data. Below is the shear exponent formula used in this calculation.

$$u_{100} = u_{50} \left( \frac{100}{50} \right)^{\left( \frac{1}{7} \right)}$$

This can be used to create a data bin that gives the frequency of a certain wind speed as seen in Figure 3. This is then compared to a Rayleigh's distribution to examine the similarities in the probability functions used.

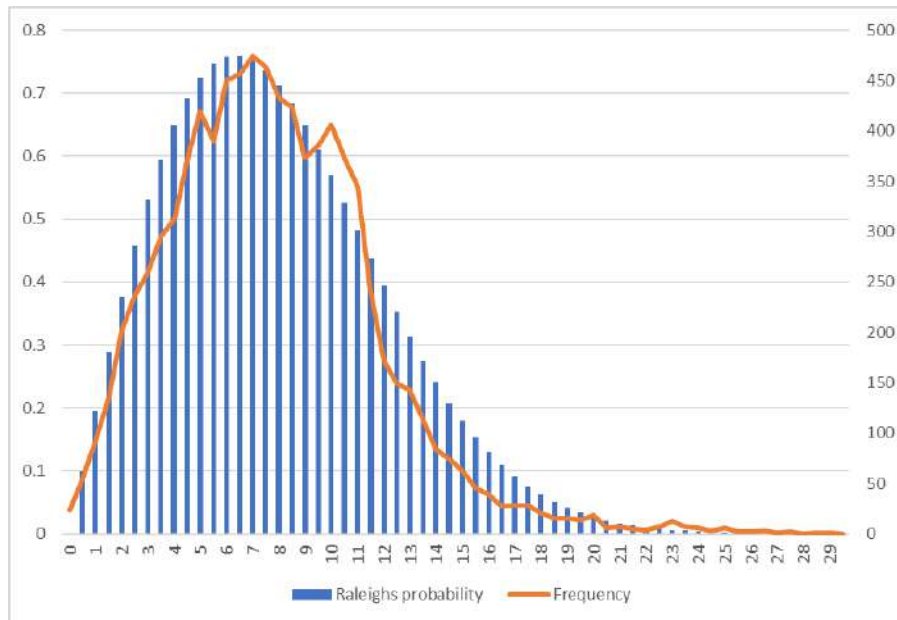


Figure 3: Wind Speed vs Frequency & Rayleigh's Probability

| Month     | Monthly Averages | SD Monthly  | TI Monthly  |
|-----------|------------------|-------------|-------------|
| January   | 8.724117627      | 4.986775221 | 0.571607976 |
| February  | 8.271125579      | 3.783380934 | 0.457420323 |
| March     | 8.522650971      | 4.395417916 | 0.515733653 |
| April     | 7.843865939      | 3.78915406  | 0.483072262 |
| May       | 7.523197683      | 3.357934335 | 0.446344025 |
| June      | 7.301589307      | 3.84789964  | 0.526994806 |
| July      | 6.802734755      | 2.89340125  | 0.425329129 |
| August    | 7.413263608      | 3.254485687 | 0.439008493 |
| September | 7.658624254      | 2.939639027 | 0.383833823 |
| October   | 8.727812766      | 3.604253226 | 0.412961795 |
| November  | 9.520103839      | 4.70460034  | 0.494175318 |
| December  | 8.099966702      | 4.008365443 | 0.494861965 |

Table 1: Wind Speed Calculations for 100M

| Annual Average | SD yearly   | TI yearly   |
|----------------|-------------|-------------|
| 8.034087752    | 3.911488303 | 0.486861536 |

Table 2: Annual Average, Standard Deviation, Turbulence Intensity Calculations for 100M

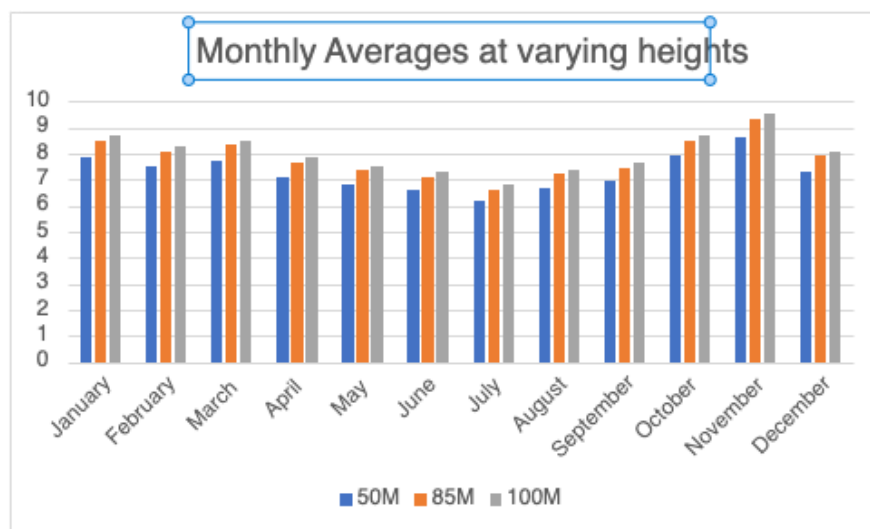


Figure 4: Monthly Averages for 50, 85 and 100M

## Turbine Selection

The turbine that provided a capacity factor of 42% at height of 100M was the Vestas turbine (V 100/1800), rated at 1800 kW with a rotor diameter of 100M. The below power curve is for V100/1800, with a polynomial estimation in Figure 6.

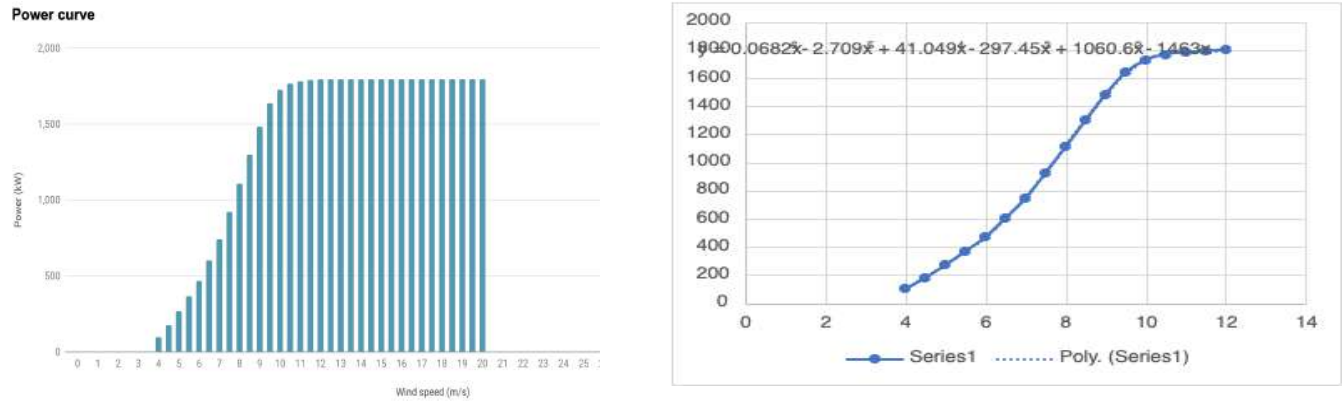


Figure 5 & 6: Power curve of V100/1.8 and Polynomial approximation of the 6th order

## Environmental Impact

It is standard nowadays to study the impact any new construction project will have on its surrounding environment, along with following necessary regulations and permits. This should be priority number one when choosing a site for a wind farm. Wildlife disturbances, historical or culturally sensitive sites, etc are all issues that need to be researched when picking a site for a new wind farm.

According to the State Historical Society of North Dakota, there are no historical sites within a 50 mile radius of Hettinger, ND. We also took into consideration that North Dakota is home to five Native American tribes when choosing our wind farm location. Our site also does not fall within any Indian reservations or culturally sensitive sites. Since the area is rural farm land and because of the points mentioned above, impact to viewshed will be minimal.

Because of the importance and value of renewable energies, there are many government sponsored studies that look at environmental impacts of all forms in renewable energies, including wind. Tethys, developed by the Pacific Northwest National Laboratory, has done most of the heavy lifting for us already in terms of environmental impact research. With North Dakota, the main species of animals to take into account are the Whooping crane and waterfowl. Waterfowl breeding locations are also crucial habitats for many wetland-dependent birds in North America, so the majority of predicted breeding areas were excluded as usable land in Figure X below. Tether also excluded whooping crane stopover sites as usable land, along with a

1.6k km buffer area around the Missouri and Red Rivers which are important migratory corridors for birds and bats. Thus, the low impact area graph was created in Figure 7 which shows that Hettinger, represented by the red dot, is in a low impact area.

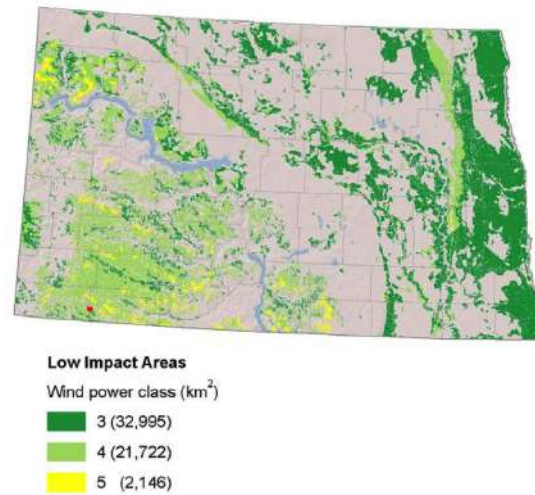


Figure 7: Low impact areas for wind development in North Dakota.

### Permits

According to Hettinger's zoning ordinance, our proposed location falls within the agriculture district. Within the agriculture district, a permitted use of land is for alternative energy production devices, such as wind turbines. We plan to pay the landowner \$40,000 a year to rent the land. Because of this supplemental income we'll be offering to the owner, we anticipate being able to partner with the landowner with minimal pushback. The landowner will also still be able to use the land for agriculture purposes.

Some additional requirements our find warm would have to follow are as follows:

- The minimum lot area with an Agricultural district shall be 5 acres
- Minimum front, rear and side yard setback shall be 50 feet from the property

### Wind Farm Expenses

#### Turbine Distribution & Direction

To provide maximum power output by reducing discrepancies caused by wake effect. To get optimal energy production, the turbines are placed, 3.5 times the rotor diameter ( $r_d$ ), perpendicular to wind direction and 7 times the  $r_d$  in the direction of the wind. With our  $r_d$  as 100m, the placement is as seen in Figure 8. This allows for a wake effect at 9.3 % for the back row.

$$\begin{aligned}
 \text{Total Energy generation} &= 3 * 6780 + (1 - X_{Wake}) * 2 * 6780 \\
 &= 32800 \text{ MWh}
 \end{aligned}$$

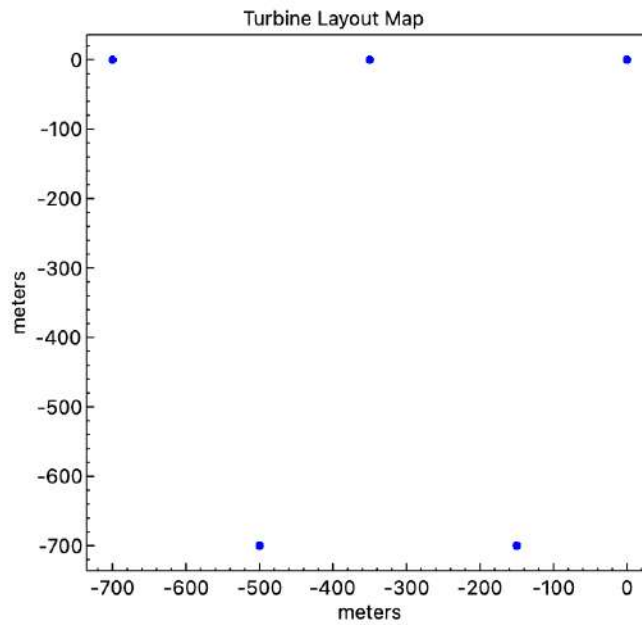


Figure 8: Turbine placement



Figure 9: Topographic representation

The satellite image shows the area covered by the turbine but we chose to buy 121 acres of land which is close to 400,000 meter squared. This is also only 500m away from a substation.

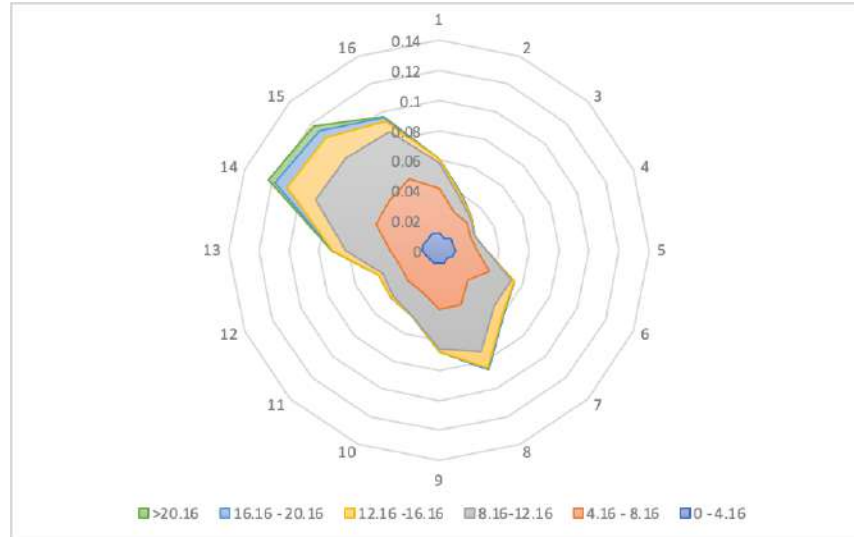


Figure 10: Wind Rose for turbine direction placement

The wind rose shows the direction of the wind speeds, with greater speeds coming mostly from the North-West. This helps to angle the wind turbine in such a direction to have greater efficiency.

### Installation Costs

The installations costs are for the V100/1.8, with a 1800 kW rated power, 100M hub height, 100M  $r_d$  and a capacity factor of 43%. The below pie chart is a total representation of the CapEx or Capital Expenditure for the turbine. This would be scaled to show the estimate of 5 turbines in the wind farm.

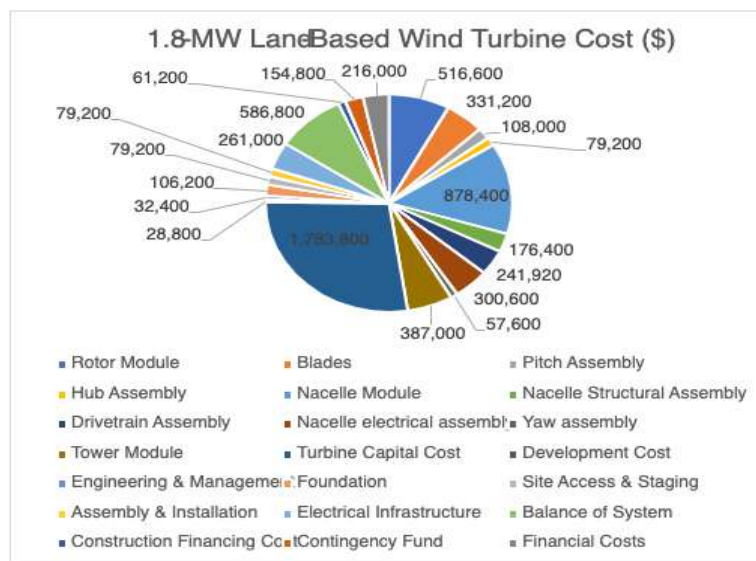


Figure 11: Total Installation costs for 1.8 MW turbine

## O&M Expenses

The average wind project has an O&M cost of \$0.01 to \$0.02 per kWh produced. A wind turbine has an energy production of 6780 MWh/yr and with \$0.01 per kWh, the cost per turbine is \$77,400 per year.

## Levelized Cost of Energy (LCOE)

The LCOE or the Levelized Cost of Energy is a measure of the cost of energy produced. It helps in the comparison between energy systems.

$$LCOE = \frac{(CapEx * FCR) + OpEx}{AEP_{net}/1000}$$

$$LCOE = \frac{(2,585,000 * 6.5\%) + 77400}{3766.7/1000}$$

$$LCOE = 33.80 \text{ \$/MWh}$$

## Losses

There are various losses as seen below that account to 15%. Losses include wake, availability, electricity, turbine performance, environmental and curtailment.

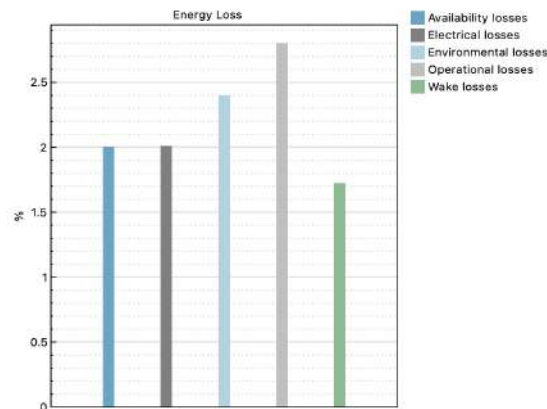


Figure 12: Losses

## Tax Expenses

We calculated an expected property tax of \$57,600 per year. This is calculated using the given value of \$6400 per MW nameplate capacity. The state tax is estimated to be \$164,000 per year in the state of North Dakota with a \$6400 per MW for 5 turbines.

To help offset some of these taxes, there is a federal tax credit. As of Dec. 31, 2021, the federal tax credit is 2.5 cents/KWh. In total this tax credit would give us \$847,500 per year. For sake of simplicity and inability to predict the future, we are planning for the tax credit amount per kWh to remain the same throughout our 10 years.

## Land Leasing and Tax

The required area of land is 121 acres or  $490,000 \text{ m}^2$ . With a \$12,500 per acre, the lease per year costs \$1,512,500. The land leasing tax is approximately \$8000 per turbine and a total of \$40,000 for 5 turbines.

## Expected Energy Production

The Vestas turbine V100/1.8 selected at given wind speeds produces the energy as seen below in kWh.

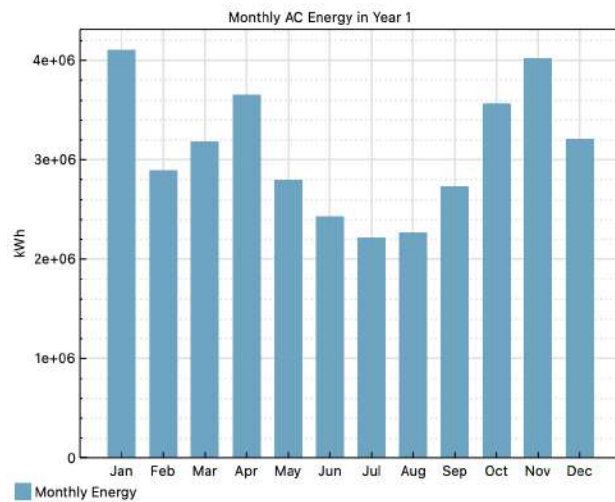


Figure 13: Monthly Energy Production

## Break-Even Analysis

*CapEx*: \$12,920,000

*Land Leasing*: 121 acres : \$12,500/acre = \$1,512,500/yr

*Property Tax* = \$6400/MW \* 9MW = \$57,600/yr

*Land Lease Tax* = \$8000/turbine \* 5 turbines = \$40,000/yr

*State Tax* = \$0.05/MW \* 328000 = \$164,000/yr

$$\text{Total Cost} = \$1,512,500 + \$57,600 + \$40,000 + \$164,000 = \$1,744,100/\text{yr}$$

$$A = \frac{(1+4\%)^{10} - 1}{4\%(1+4\%)^{10}} = 8.1109$$

$$P = A(P/A, 4\%, 10) = \$1,744,100/\text{yr} * 8.1109 = \$14,389,723$$

$$\text{Total} = \$14,389,723 + \$12,920,000 = \$27,309,723$$

$$\text{Retail (cent/kWh)} = \frac{\$27,309,723}{32,800 * 10^3 * 8.1109} = 10.265 \text{ cent/kWh}$$

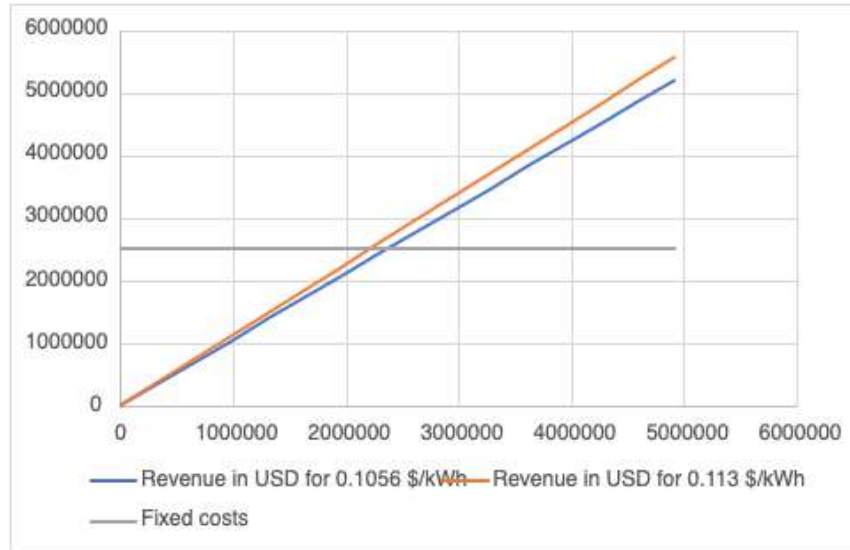


Figure 14: kWh vs Revenue earned for 10.2 cents and 11.3 cents

Thus to reach the break even point at 10 years, we would have to sell at 10.265 cents/kWh. This is 9% less than the national average which is 11.3 cents/kWh. The above breakeven is not in reference to time but the revenue at which the break in occurs for 10.265 cents and 11.3 cents.

## Summary

Our wind farm is estimated to produce 32.8 GWh annually which will be sold at 10.265 cents/kWh. This is below the national average that Americans currently pay for electricity. On top of our power being cheaper, the other benefit is that this is 100% clean, renewable energy being generated for the town of Hettinger. Our annual revenue is estimated to be \$3,366,920 along with a \$847,500 tax credit we'll receive each year. With a sum of yearly expenses calculated out to be \$27,309,723, we anticipate to break even at the 10 year mark as long as we sell our electricity at 10.265 cents/kWh. With most wind farms lasting between 20-25 years, not only do we see this as a good investment for the town of Hettinger, but we also see it as a good personal investment.

## References

Excel Sheet Calculations: [Wind\\_data\\_Hettinger.xlsx](#)

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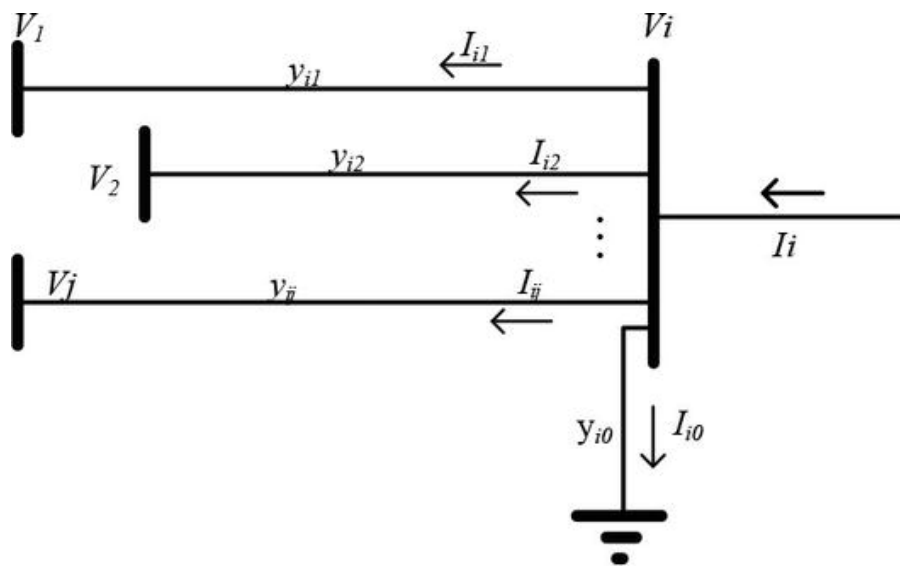
# ***ETAP Power System Analysis and Load Flow***

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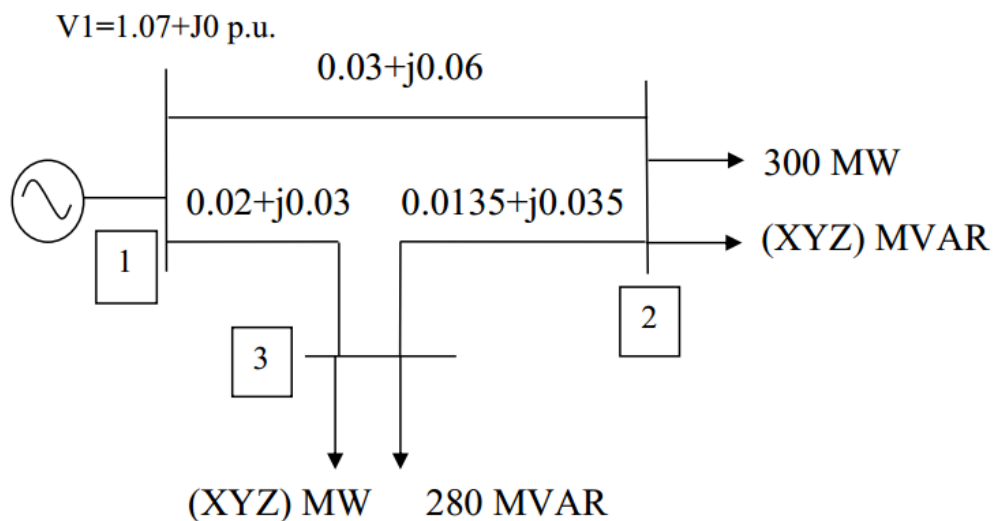
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## 1 Introduction

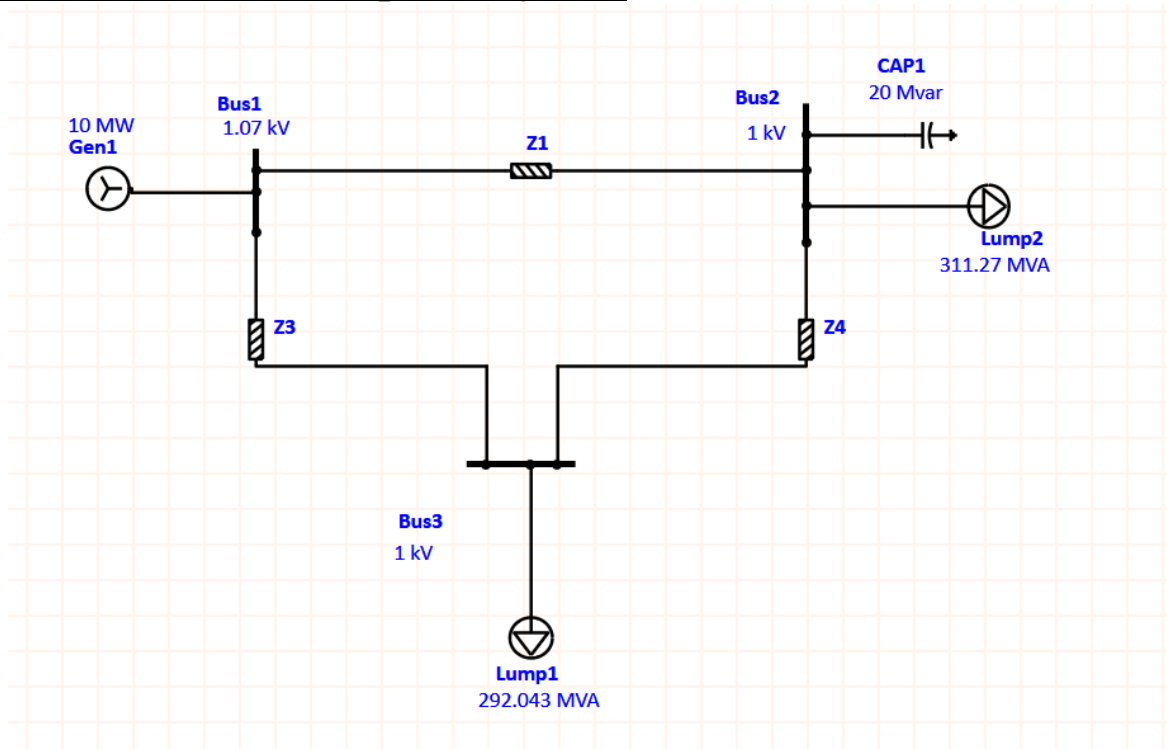
The Load Flow Analysis is a fundamental tool in interconnected power system analysis. Load Flow Analysis results play major role during the operational stages of any power system for its control and economic schedule, as well as during the expansion or design stages. The main purpose of any Load Flow Analysis is to compute precise steady-state voltages and voltage angles of all buses in the network, the reactive and real power flows into every overhead transmission line and power transformer (located at Sub-stations), under the assumption of known generation and load. Using a software such as ETAP eases the load flow calculations and analysis.



## 2 Power system model:

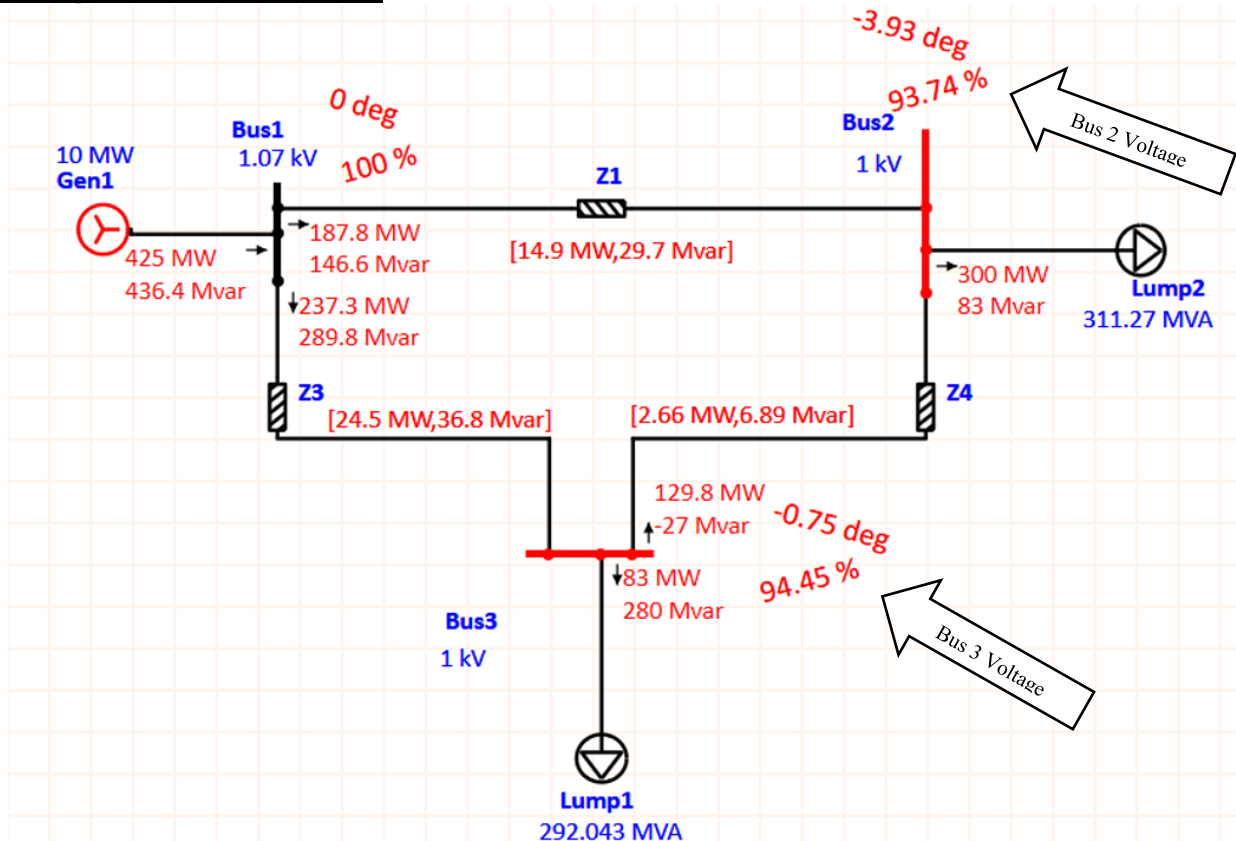


### 3 ETAP Simulation of the power system

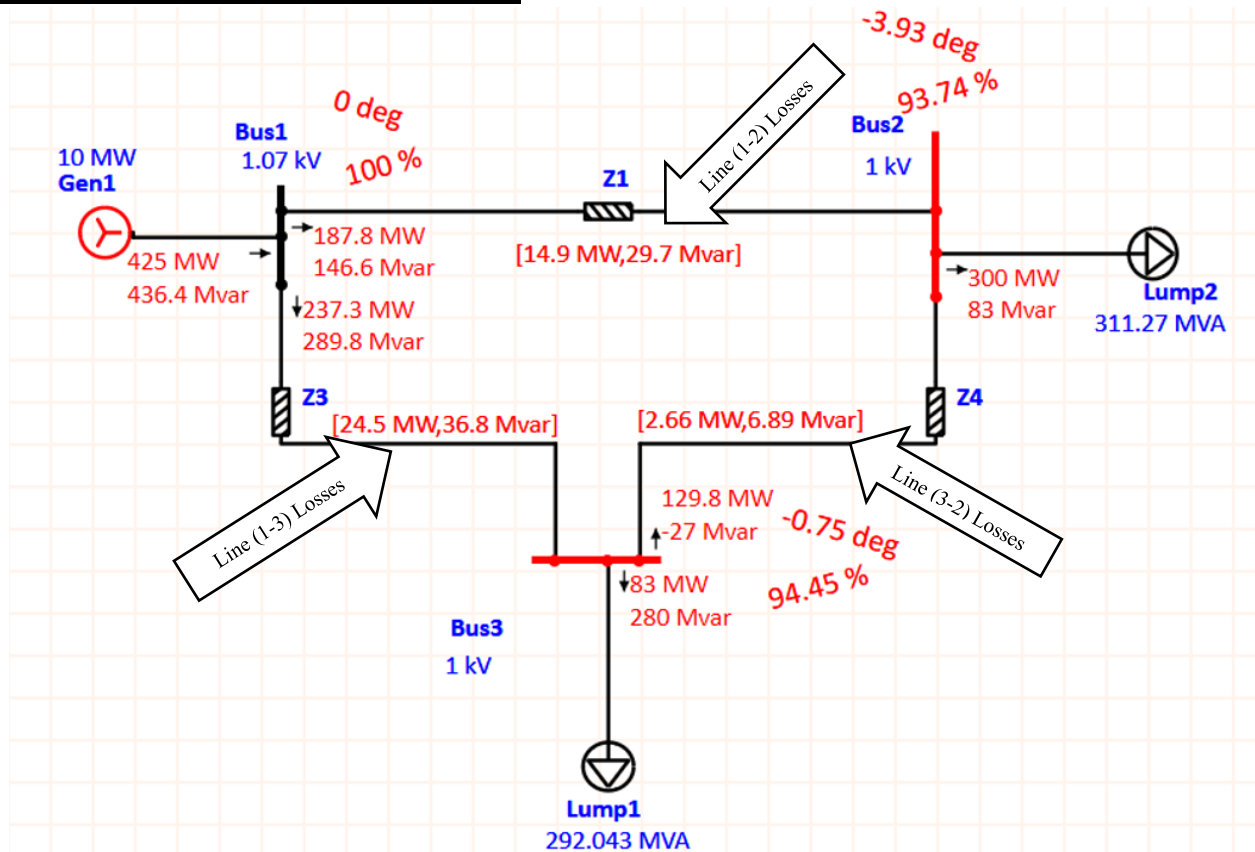


### 4 Snapshots of the ETAP

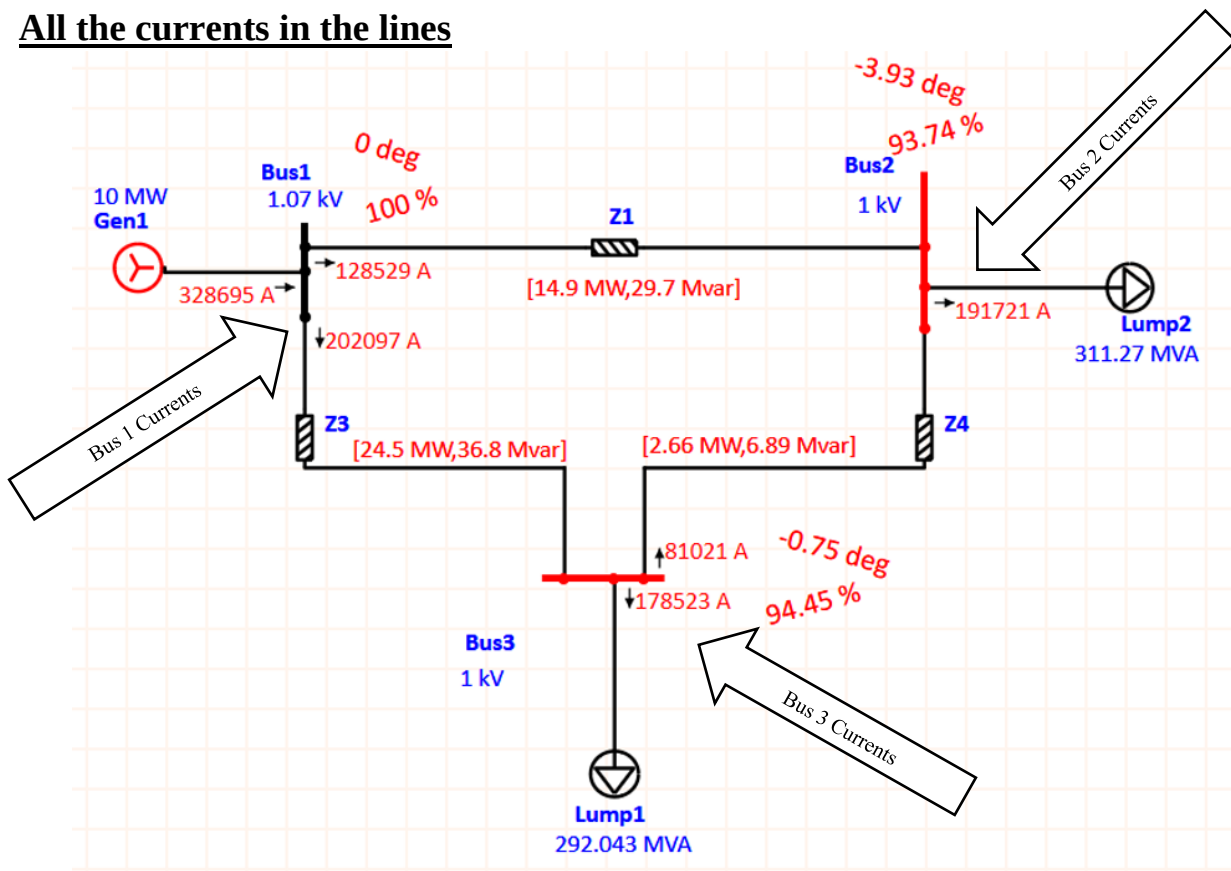
#### 4.1 Voltages at buses 2 and 3



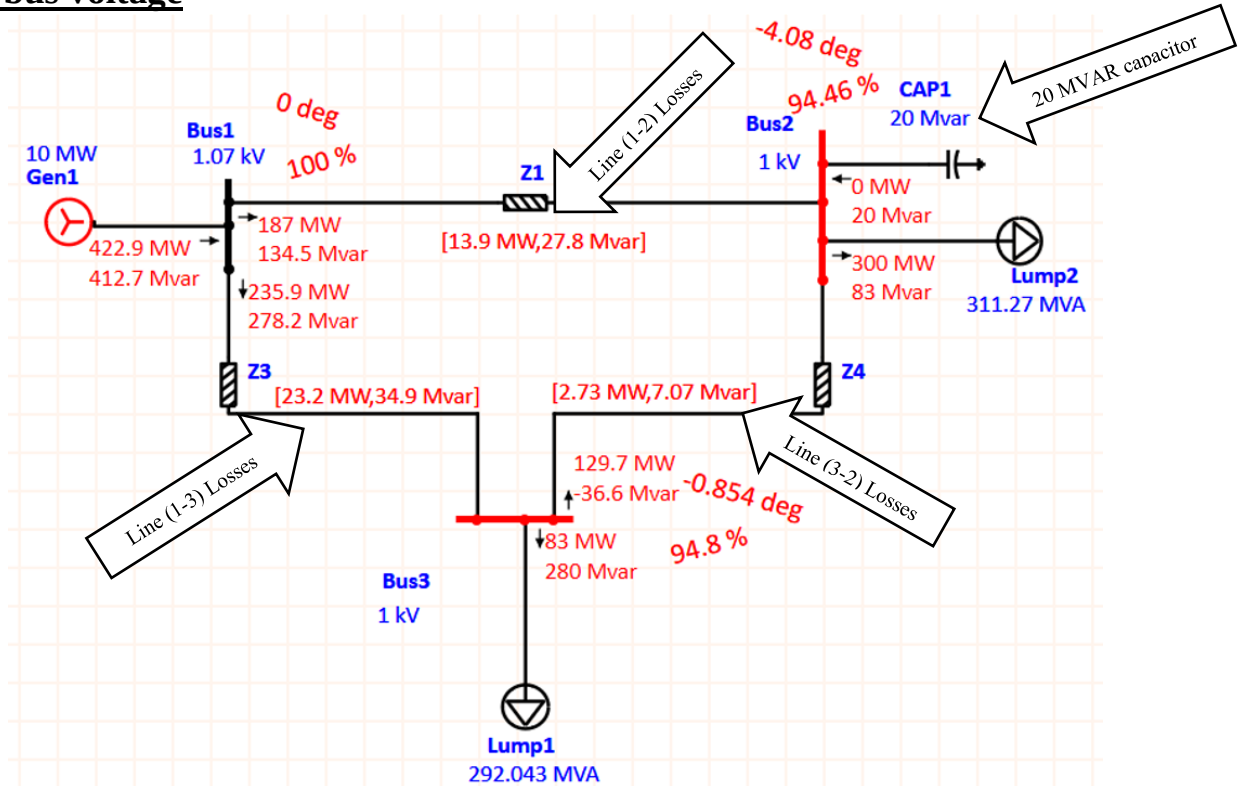
## 4.2 All active and reactive line losses.



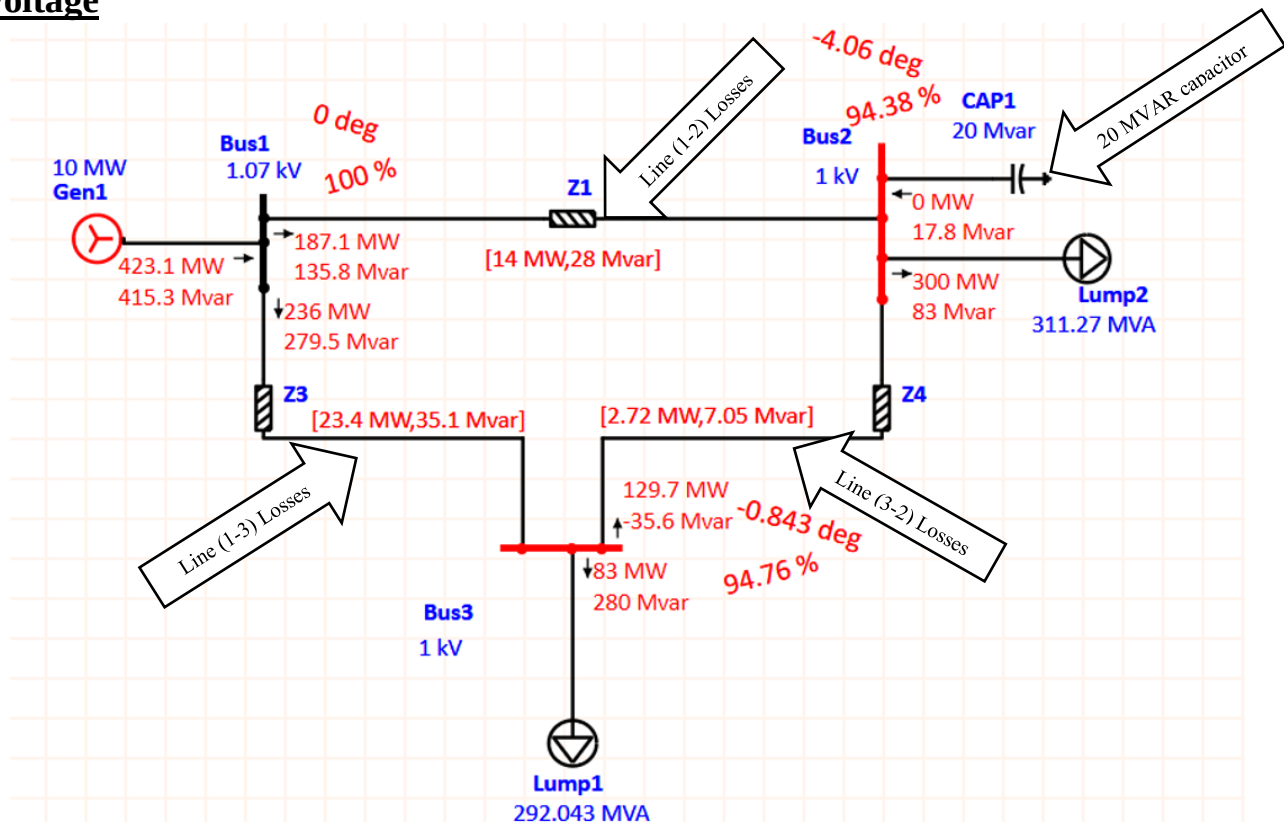
## 4.3 All the currents in the lines



#### 4.4 The effect of 20 MVAR capacitor bank at bus 2 on the line losses, with voltage rating same bus voltage



#### 4.5 The effect of 20 MVAR capacitor bank at bus 2 on the line losses, with 1kv rating bus voltage



#### 4.6 The effect of the voltage magnitude and angle on the active and reactive power flow

From the simulation results obtained, it is easily shown that the active and reactive power flow from the higher voltage and angle (Ex: at bus (1)  $1.07 < 0$ ) to the lower voltage and angle (at bus (2)  $0.9374 < -3.93$ ). The following equations used to find the active and reactive power flow from each bus and how they are related to the bus voltage:

$$V_i^{[k+1]} = \frac{\frac{P_i^{[sch]} - jQ_i^{[sch]}}{V_i^{*[k]}} - \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]}}{Y_{ii}}$$

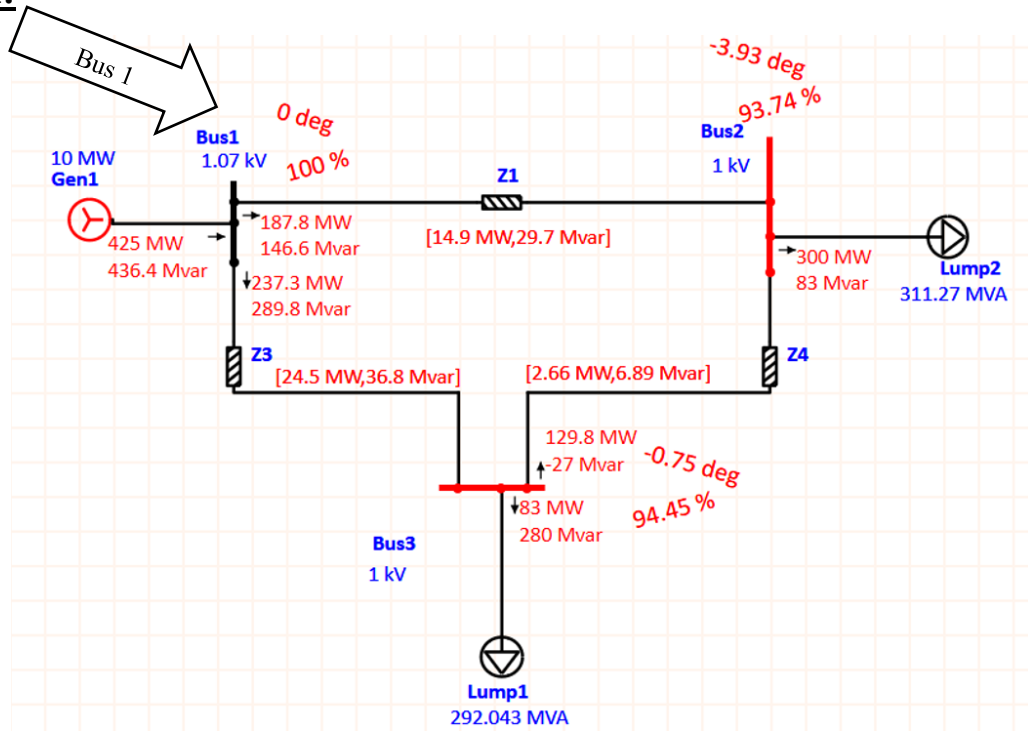
$$P_i^{[k+1]} = \Re \left\{ V_i^{*[k]} \left[ V_i^{[k]} Y_{ii} + \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]} \right] \right\}$$

$$Q_i^{[k+1]} = -\Im \left\{ V_i^{*[k]} \left[ V_i^{[k]} Y_{ii} + \sum_{j=1, j \neq i}^n Y_{ij} V_j^{[k]} \right] \right\}$$

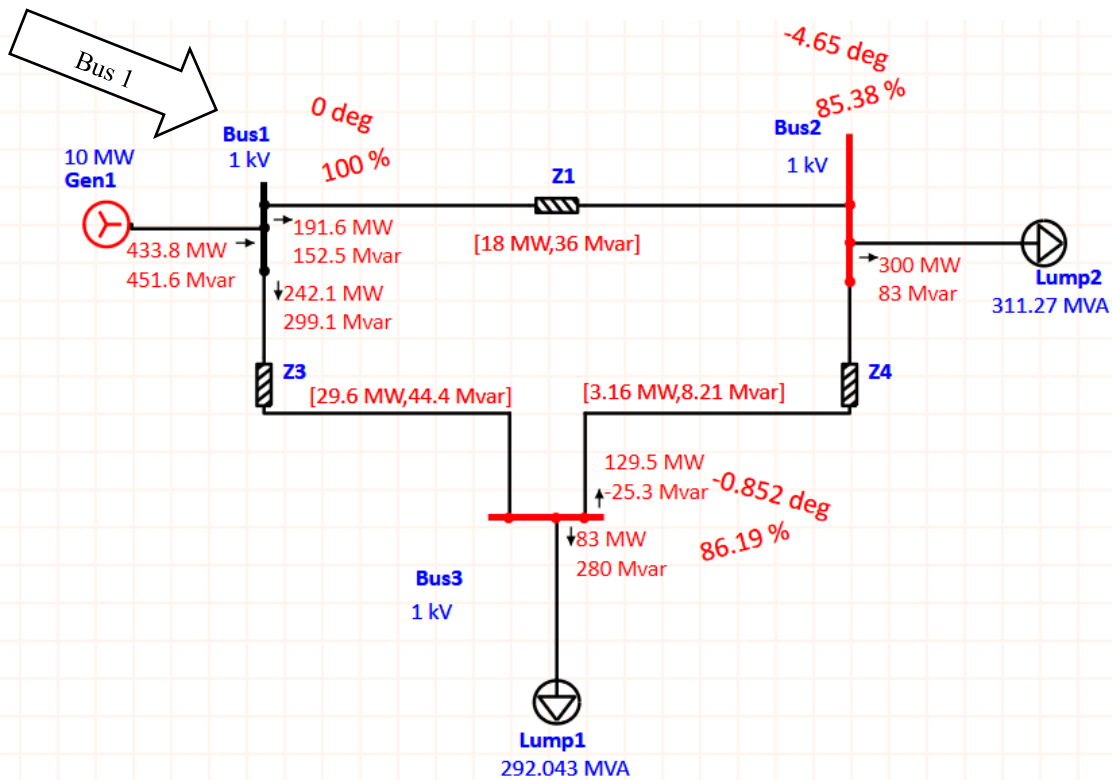
| Case 1                    |                  | Case 2                    |                   |
|---------------------------|------------------|---------------------------|-------------------|
| Bus 1 Voltage and angle   | $1.07 < 0$       | Bus 1 Voltage and angle   | $1.00 < 0$        |
| Bus 2 Voltage and angle   | $0.9374 < -3.93$ | Bus 2 Voltage and angle   | $0.8538 < -4.65$  |
| Bus 3 Voltage and angle   | $0.9445 < -0.75$ | Bus 3 Voltage and angle   | $0.8619 < -0.852$ |
| Active power flow (1-2)   | 187.8 MW         | Active power flow (1-2)   | 191.6 MW (↑)      |
| Reactive power flow (1-2) | 146.6 Mvar       | Reactive power flow (1-2) | 152.5 Mvar (↑)    |
| Active power flow (1-3)   | 237.3 MW         | Active power flow (1-3)   | 242.1 MW (↑)      |
| Reactive power flow (1-3) | 289.8 Mvar       | Reactive power flow (1-3) | 299.1 Mvar (↑)    |
| Active power flow (2-3)   | 129.8 MW         | Active power flow (2-3)   | 129.5 MW (↑)      |
| Reactive power flow (2-3) | -27 Mvar         | Reactive power flow (2-3) | -25.3 Mvar (↑)    |

In case we are comparing two different systems with changing same Bus Voltage, as we reduce the difference between Bus Voltages and the voltage angles, the active and reactive power flow increases. The following screenshots shows this concept.

#### 4.6.1 Case 1:



#### 4.6.2 Case 2:



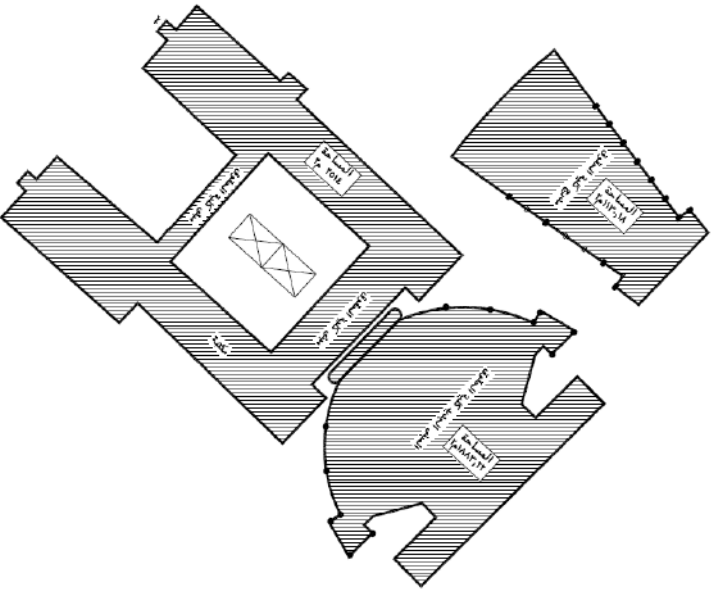
# PV On-Grid System

## Report Contents:

1. University Building Identification.
2. Radiation Data.
3. RETScreen Report.
4. PV Module and Inverter Selection and Characteristics.
5. System Sizing and Design Calculations.
6. PV System Summary.
7. PVsyst Report.

.....

## University Building Identification.

|                                                                                                                                                                                   |                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>University Name:</b>                                                                                                                                                           | Ain Shams University                                                                                                                                      |
| <b>Location (City Name and GPS Coordinates)</b>                                                                                                                                   | Cairo.                                                                                                                                                    |
| <b>Title of Building* <sup>1</sup> where the EE/RE opportunities will be implemented:</b>                                                                                         | Faculty of Law.                                                                                                                                           |
| <b>Building Address:</b>                                                                                                                                                          | Campus (1) - Ain Shams University - El-Khalifa El-Mamoun El-Abbasyia, Cairo, Egypt                                                                        |
| <b>Estimated Annual Building Energy Consumption (MWh): Contribution of Building's Energy Consumption to Overall Energy Consumption of the Faculty and University: (ratio x:y)</b> | <hr/> <p>Building Energy Consumption / Faculty Energy Consumption = 629 MWh/Year</p> <p>Approximate Faculty/ University Campus 1 Consumption % = 12 %</p> |
| <b>Building Location:</b>                                                                                                                                                         |  <p>A2. Faculty of Law.</p>                                            |

|                                                                                                                    |                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                    | <div><p>Faculty of Law - Circled in RED</p></div> |
| Building Footprint Area (m <sup>2</sup> )                                                                          | Main Building == > 4397 m2 & Lecture Hall ==> 1,130.18 m2                                                                           |
| Building Number of Floors                                                                                          | Basement & Four Floors                                                                                                              |
| Building Functions, i.e. Admin, Faculty, Department, Lectures, Lab, Library, Workshop, Student Services,..... etc. | Admin, Faculty, Department, Lectures, Lab, Library, Workshop                                                                        |

.....

## Radiation data

Based on PVGIS Tool [Photovoltaic Geographical Information System]

Fixed mounting System

**Cursor:** 30.095, 31.314  
**Selected:** 30.077, 31.286  
Elevation (m): 29  
PVGIS ver. 5.2

**Use terrain shadows:**  
☒ Calculated horizon  
☐ Upload horizon file  
[Switch to version 5.1](#)

[Download CSV](#)  
Choose File

[Download JSON](#)  
No file chosen

GRID CONNECTED

TRACKING PV

OFF-GRID

MONTHLY DATA

DAILY DATA

HOURLY DATA

TMY

 **PERFORMANCE OF GRID-CONNECTED PV** 

Solar radiation database\*  
PV technology\*  
Installed peak PV power [kWp]\*  
System loss [%]\*

PVGIS-SARAH2  
Crystalline silicon  
207  
15

**Fixed mounting options**  
Mounting position \*  
Slope [°]\*  
Azimuth [°]\*  
☐ **PV electricity price**  
PV system cost (your currency)  
Interest [%/year]  
Lifetime [years]

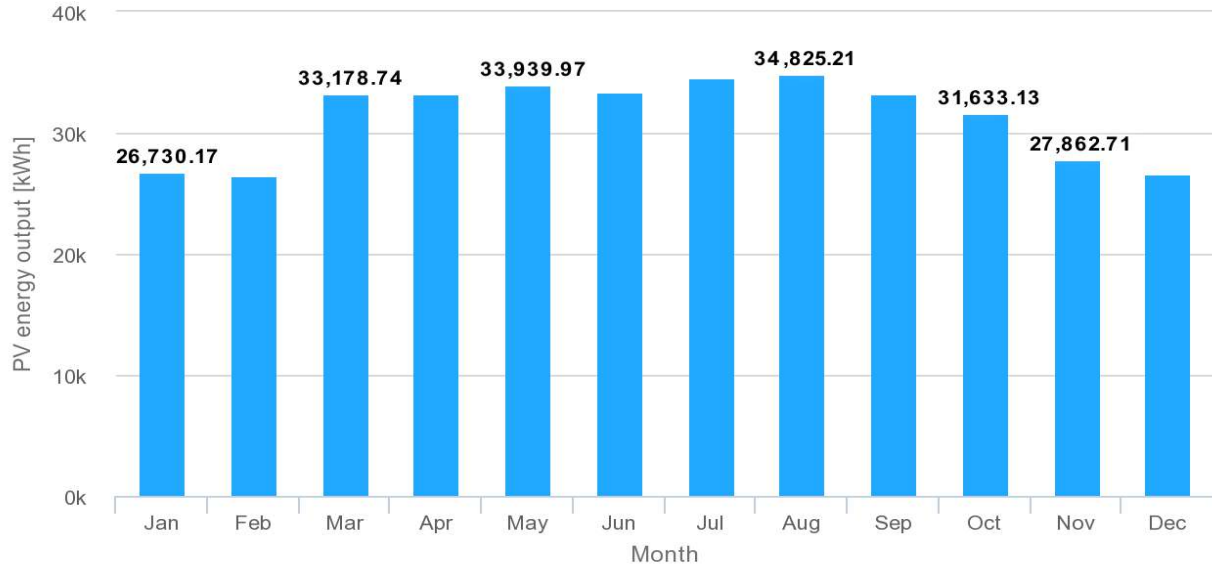
Free-standing  
Optimize slope  
☒ Optimize slope and azimuth

### Summary

| Provided inputs:                                   |                     |
|----------------------------------------------------|---------------------|
| Location [Lat/Lon]:                                | 30.077,31.286       |
| Horizon:                                           | Calculated          |
| Database used:                                     | PVGIS-SARAH2        |
| PV technology:                                     | Crystalline silicon |
| PV installed [kWp]:                                | 207                 |
| System loss [%]:                                   | 15                  |
| Simulation outputs:                                |                     |
| Slope angle [°]:                                   | 28 (opt)            |
| Azimuth angle [°]:                                 | 5 (opt)             |
| Yearly PV energy production [kWh]:                 | 375910.27           |
| Yearly in-plane irradiation [kWh/m <sup>2</sup> ]: | 2429.15             |
| Year-to-year variability [kWh]:                    | 3446.61             |
| Changes in output due to:                          |                     |
| Angle of incidence [%]:                            | -2.48               |
| Spectral effects [%]:                              | 0.05                |
| Temperature and low irradiance [%]:                | -9.85               |
| Total loss [%]:                                    | -25.24              |

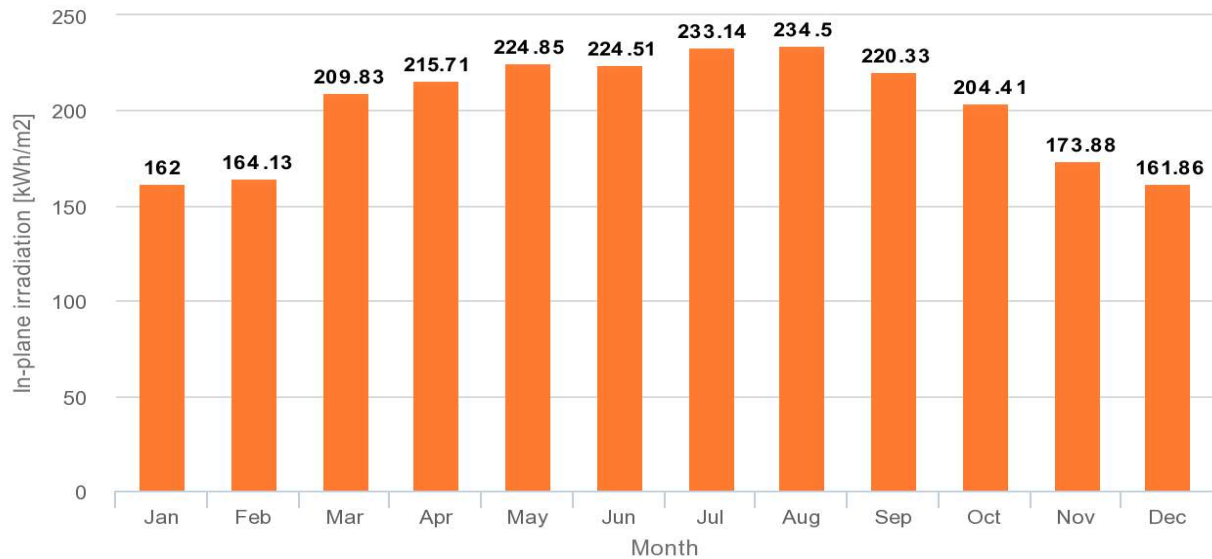
### Monthly energy output from fix-angle PV system

(C) PVGIS, 2023



### Monthly in-plane irradiation for fixed angle

(C) PVGIS, 2023



| Mounting | Yearly PV energy production [kWh] |
|----------|-----------------------------------|
| Fixed    | 375910.27                         |

.....

# RETScreen Report

## Feasibility report

Ain Shams University

200 kW



Power plant - Photovoltaic

Prepared for:

Graduation Project

Prepared by:

Mohamed Tawheed

## Executive summary

This report was prepared using the RETScreen Clean Energy Management Software. The key findings of this analysis are presented

Target

|               | Electricity exported to<br>grid<br>MWh | Electricity export<br>revenue<br>\$ | GHG emission<br>reduction<br>tCO <sub>2</sub> |
|---------------|----------------------------------------|-------------------------------------|-----------------------------------------------|
| Proposed case | 378                                    | 37,760                              | 183                                           |

# Site reference conditions

Climate data location
Egypt - Cairo H.Q.

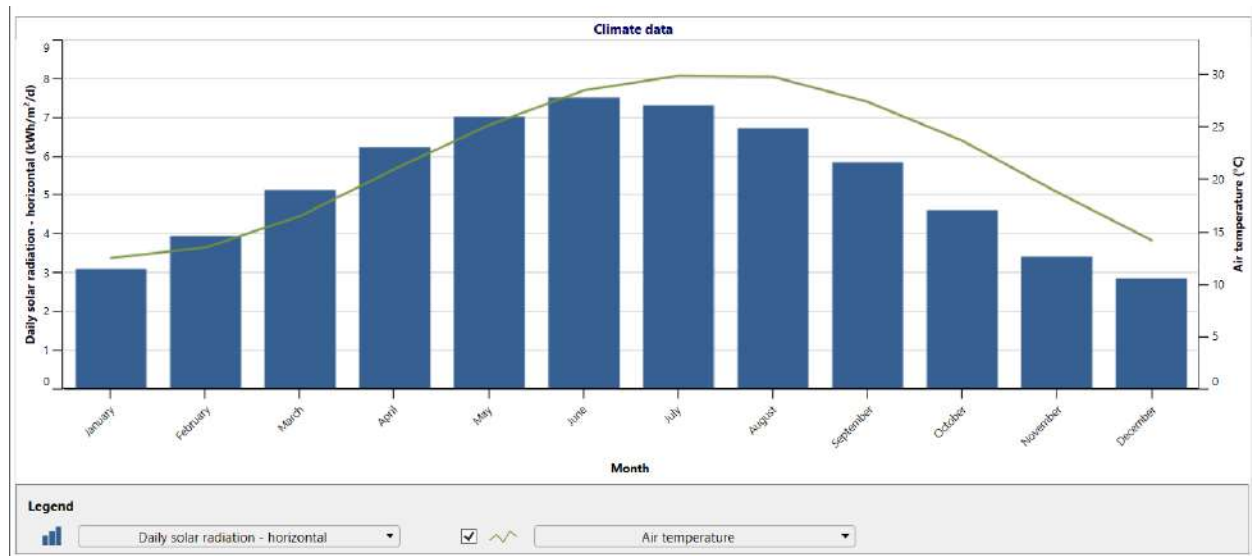
Facility location
Egypt - Cairo - Qesm El Waily

- Legend**
- Facility location
  - Climate data location



|                             | Unit | Climate data location | Facility location | Source       |
|-----------------------------|------|-----------------------|-------------------|--------------|
| Latitude                    |      | 30.1                  | 30.1              |              |
| Longitude                   |      | 31.3                  | 31.3              |              |
| Climate zone                |      | 2B - Hot - Dry        |                   |              |
| Elevation                   | m    | 26                    | 28                | NASA         |
| Heating design temperature  | °C   | 7.0                   |                   | Ground - Map |
| Cooling design temperature  | °C   | 34.7                  |                   | NASA         |
| Earth temperature amplitude | °C   | 21.4                  |                   | NASA         |

| Month         | Air temperature<br>°C | Relative humidity<br>% | Precipitation<br>mm | Daily solar radiation - horizontal<br>kWh/m <sup>2</sup> /d | Atmospheric pressure<br>kPa | Wind speed<br>m/s | Earth temperature<br>°C | Heating degree-days<br>18 °C<br>°C-d | Cooling degree-days<br>10 °C<br>°C-d |
|---------------|-----------------------|------------------------|---------------------|-------------------------------------------------------------|-----------------------------|-------------------|-------------------------|--------------------------------------|--------------------------------------|
| January       | 12.5                  | 60.5%                  | 8.37                | 3.09                                                        | 100.9                       | 3.6               | 12.5                    | 171                                  | 78                                   |
| February      | 13.5                  | 54.4%                  | 7.00                | 3.94                                                        | 100.8                       | 3.8               | 13.7                    | 126                                  | 98                                   |
| March         | 16.5                  | 48.1%                  | 7.44                | 5.13                                                        | 100.6                       | 4.0               | 17.4                    | 47                                   | 202                                  |
| April         | 21.0                  | 39.0%                  | 3.00                | 6.23                                                        | 100.4                       | 4.2               | 22.5                    | 0                                    | 330                                  |
| May           | 25.2                  | 35.2%                  | 0.93                | 7.02                                                        | 100.3                       | 4.3               | 27.4                    | 0                                    | 471                                  |
| June          | 28.5                  | 34.8%                  | 0.00                | 7.51                                                        | 100.1                       | 4.4               | 31.0                    | 0                                    | 555                                  |
| July          | 29.9                  | 37.0%                  | 0.00                | 7.31                                                        | 99.8                        | 4.0               | 32.4                    | 0                                    | 617                                  |
| August        | 29.8                  | 39.9%                  | 0.00                | 6.72                                                        | 99.9                        | 3.7               | 32.1                    | 0                                    | 614                                  |
| September     | 27.4                  | 45.0%                  | 0.30                | 5.84                                                        | 100.2                       | 3.9               | 29.2                    | 0                                    | 522                                  |
| October       | 23.7                  | 51.0%                  | 1.86                | 4.61                                                        | 100.5                       | 3.7               | 25.0                    | 0                                    | 425                                  |
| November      | 18.8                  | 56.0%                  | 4.50                | 3.41                                                        | 100.7                       | 3.6               | 19.3                    | 0                                    | 264                                  |
| December      | 14.2                  | 60.3%                  | 6.20                | 2.85                                                        | 100.9                       | 3.6               | 14.1                    | 118                                  | 130                                  |
| <b>Annual</b> | <b>21.8</b>           | <b>46.7%</b>           | <b>39.60</b>        | <b>5.31</b>                                                 | <b>100.4</b>                | <b>3.9</b>        | <b>23.1</b>             | <b>461</b>                           | <b>4,305</b>                         |
| Source        | NASA                  | NASA                   | NASA                | Ground                                                      | NASA                        | NASA              | NASA                    | NASA                                 | NASA                                 |
| Measured at   |                       |                        |                     |                                                             | m                           | 10                | 0                       |                                      |                                      |



Power plant

Photovoltaic  
200 kW



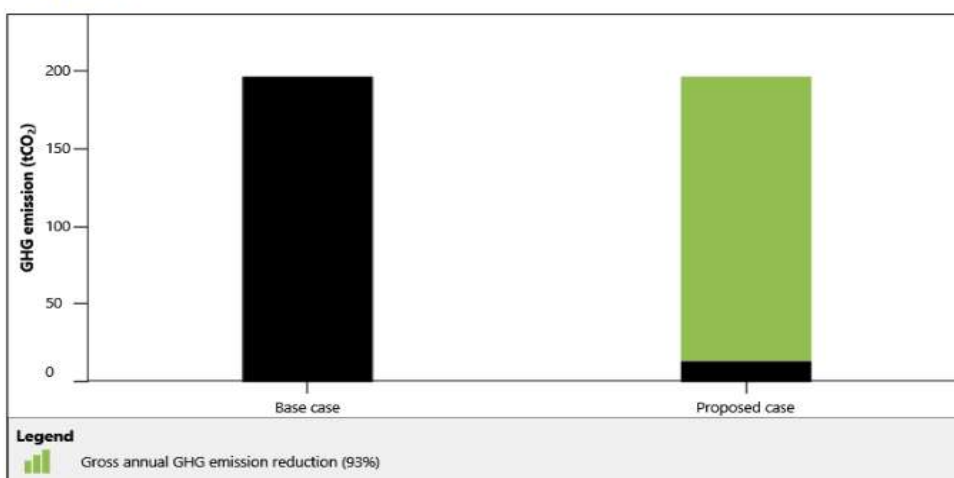
| Photovoltaic - 200 kW |     |     |
|-----------------------|-----|-----|
| Capacity              | 207 | kW  |
| Electricity           | 378 | MWh |

Target

Summary

|               | Electricity exported to<br>grid<br>MWh | Electricity export<br>revenue<br>\$ | GHG emission<br>reduction<br>tCO <sub>2</sub> |
|---------------|----------------------------------------|-------------------------------------|-----------------------------------------------|
| Proposed case | 378                                    | 37,760                              | 183                                           |

## GHG emission



## GHG equivalence



183.1 tCO<sub>2</sub> is equivalent to 78,689  
Litres of gasoline not consumed

### GHG emission

|                                            |              |                        |
|--------------------------------------------|--------------|------------------------|
| Base case                                  | 196.9        | tCO <sub>2</sub>       |
| Proposed case                              | 13.8         | tCO <sub>2</sub>       |
| <b>Gross annual GHG emission reduction</b> | <b>183.1</b> | <b>tCO<sub>2</sub></b> |

## Financial viability

### Financial parameters

#### General

|                   |    |    |
|-------------------|----|----|
| Inflation rate    | %  | 2% |
| Discount rate     | %  | 9% |
| Reinvestment rate | %  | 9% |
| Project life      | yr | 25 |

#### Finance

|                    |       |         |
|--------------------|-------|---------|
| Debt ratio         | %     | 70%     |
| Debt               | \$    | 144,900 |
| Equity             | \$    | 62,100  |
| Debt interest rate | %     | 7%      |
| Debt term          | yr    | 15      |
| Debt payments      | \$/yr | 15,909  |

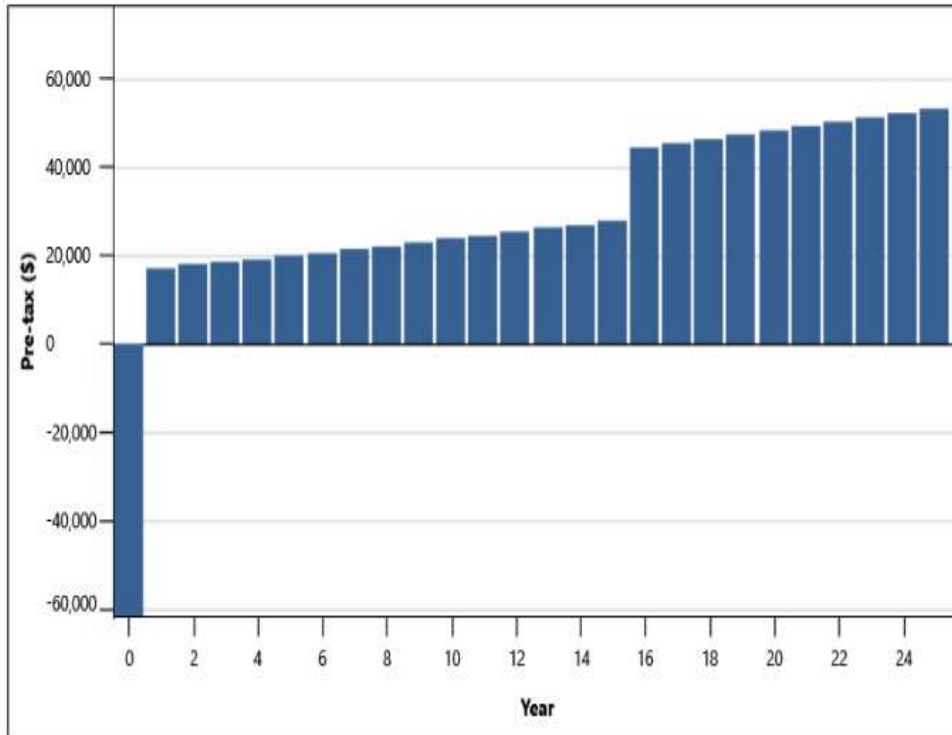
### Annual revenue

#### Electricity export revenue

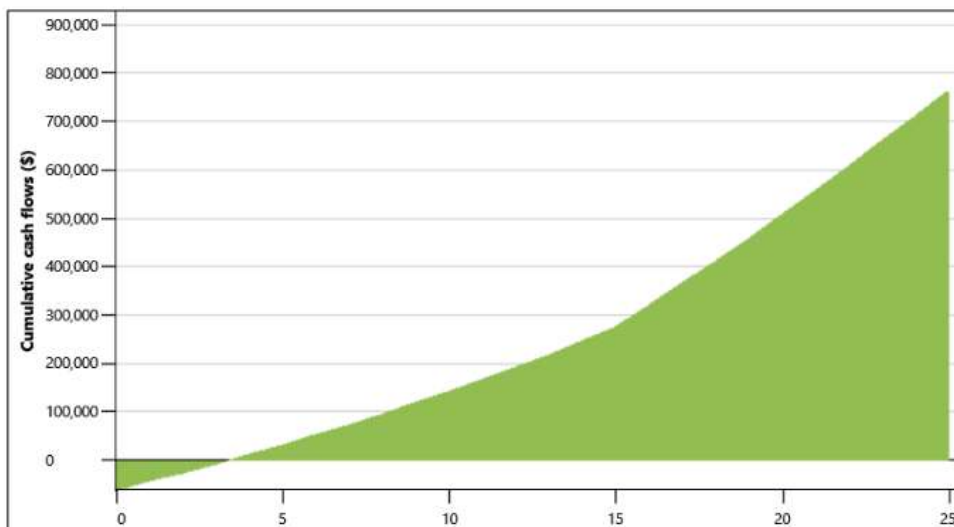
|                                    |        |        |
|------------------------------------|--------|--------|
| Electricity exported to grid       | MWh    | 378    |
| Electricity export rate            | \$/kWh | 0.10   |
| Electricity export revenue         | \$     | 37,760 |
| Electricity export escalation rate | %      | 2%     |

# Cash flow

Annual



Cumulative



.....

## PV SUNTECH module Specifications [Data Sheet]

### Electrical Characteristics

| STC                             | STPXXXS-C72/Vmh  |        |        |        |        |
|---------------------------------|------------------|--------|--------|--------|--------|
| Maximum Power at STC (Pmax)     | 550W             | 545W   | 540W   | 535W   | 530W   |
| Optimum Operating Voltage (Vmp) | 42.05V           | 41.87V | 41.75V | 41.57V | 41.39V |
| Optimum Operating Current (Imp) | 13.08A           | 13.02A | 12.94A | 12.87A | 12.81A |
| Open Circuit Voltage (Voc)      | 49.88V           | 49.69V | 49.54V | 49.39V | 49.24V |
| Short Circuit Current (Isc)     | 14.01A           | 13.96A | 13.89A | 13.83A | 13.76A |
| Module Efficiency               | 21.3%            | 21.1%  | 20.9%  | 20.7%  | 20.5%  |
| Operating Module Temperature    | -40 °C to +85 °C |        |        |        |        |
| Maximum System Voltage          | 1500 V DC (IEC)  |        |        |        |        |
| Maximum Series Fuse Rating      | 25 A             |        |        |        |        |
| Power Tolerance                 | 0/+5 W           |        |        |        |        |

STC: Irradiance 1000 W/m<sup>2</sup>, module temperature 25 °C, AM=1.5;  
Tolerance of Pmax is within +/- 3% ;  
For tracker installation, please turn to Suntech for mechanical load information.

| NMOT                            | STPXXXS-C72/Vmh |        |        |        |        |
|---------------------------------|-----------------|--------|--------|--------|--------|
| Maximum Power at NMOT (Pmax)    | 415.0W          | 411.5W | 408.0W | 404.3W | 400.6W |
| Optimum Operating Voltage (Vmp) | 38.9V           | 38.7V  | 38.6V  | 38.4V  | 38.2V  |
| Optimum Operating Current (Imp) | 10.67A          | 10.63A | 10.58A | 10.53A | 10.47A |
| Open Circuit Voltage (Voc)      | 46.9V           | 46.7V  | 46.5V  | 46.4V  | 46.3V  |
| Short Circuit Current (Isc)     | 11.22A          | 11.18A | 11.13A | 11.08A | 11.02A |

NMOT: Irradiance 800 W/m<sup>2</sup>, ambient temperature 20 °C, AM=1.5, wind speed 1 m/s.

### Temperature Characteristics

|                                             |            |
|---------------------------------------------|------------|
| Nominal Module Operating Temperature (NMOT) | 42 ± 2 °C  |
| Temperature Coefficient of Pmax             | -0.36%/°C  |
| Temperature Coefficient of Voc              | -0.304%/°C |
| Temperature Coefficient of Isc              | 0.050%/°C  |

### Mechanical Characteristics

|               |                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solar Cell    | Monocrystalline silicon 182 mm                                                                                                                       |
| No. of Cells  | 144 (6 × 24)                                                                                                                                         |
| Dimensions    | 2279 × 1134 × 35 mm (89.7 × 44.6 × 1.4 inches)                                                                                                       |
| Weight        | 29.1 kgs (64.2 lbs.)                                                                                                                                 |
| Front Glass   | 3.2 mm (0.126 inches)                                                                                                                                |
| Frame         | Anodized aluminium alloy                                                                                                                             |
| Junction Box  | IP68 rated (3 bypass diodes)                                                                                                                         |
| Output Cables | 4.0 mm <sup>2</sup> ,<br>Portrait: (-) 350 mm and (+) 160 mm in length<br>Landscape : (-) 1 400 mm and (+) 1400 mm in length<br>or customized length |
| Connectors    | MC4 EVO2, Cable 01S                                                                                                                                  |

### SUNTECH Module Characteristics

Maximum Power at STC ( $P_{\max}$ )= 540W.

Current at  $P_{\max}$  = 12.94 A

Voltage at  $P_{\max}$  = 41.75 V

S.C Current = 13.89 A

O.C Voltage = 49.54 V

Maximum System Voltage = 1500 V DC (IEC)

Nominal Module Operating Temperature (NMOT) =  $42 \pm 2$  °C

Dimensions = 2279 mm\* 1134 mm\* 35 mm

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# Huawei Smart String Inverter Specifications [Data Sheet]



## Smart String Inverter (SUN2000-60KTL-HV-D1-001)

| Technical Specifications            | SUN2000-60KTL-HV-D1-001 |
|-------------------------------------|-------------------------|
|                                     | <b>Efficiency</b>       |
| Max. Efficiency                     | 99.0%                   |
| European Efficiency                 | 98.8%                   |
|                                     | <b>Input</b>            |
| Max. Input Voltage                  | 1,500 V                 |
| Max. Current per MPPT               | 22 A                    |
| Max. Short Circuit Current per MPPT | 30 A                    |
| Start Voltage                       | 650 V                   |
| MPPT Operating Voltage Range        | 600 V ~ 1,450 V         |
| Rated Input Voltage                 | 1,080 V                 |
| Number of Inputs                    | 8                       |
| Number of MPP Trackers              | 4                       |
|                                     | <b>Output</b>           |
| Rated AC Active Power               | 60,000 W                |
| Max. AC Apparent Power              | 66,000 VA               |
| Max. AC Active Power (cosφ=1)       | 66,000 W                |
| Rated Output Voltage                | 800 V, 3W + PE          |
| Rated AC Grid Frequency             | 50 Hz / 60 Hz           |
| Rated Output Current                | 43.3 A                  |
| Max. Output Current                 | 48 A                    |

### Huawei – SUN2000-60KTL-HV-D1-001 Inverter Characteristics

Unit Nominal Power = 60 KW AC

DC Voltage Range MPPT = 600 V – 1450 V

Max. Input Voltage = 1500 V

Max. Current per MPPT = 22 A

Max. Short Circuit Current per MPPT = 30 A

Number of Inputs = 8

Number of MPP Trackers = 4

PF = 0.9

$\eta_{Inv} = 99 \%$

Panel Protection: IP65

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## Sizing and Design of Solar PV System

$$\text{Number of modules} = \frac{\text{Solar PV system power (KW)} * 1,000}{\text{PV module power (W)}}$$

$$\text{Number of modules} = \frac{207 * 1,000}{540} = \mathbf{384 \text{ Module}}$$

For ground installation with row spacing, 1 kW of Solar PV system needs approximately: 8-10 m<sup>2</sup>

$$\text{PV System Area} = 207 * [8 - 10] = [\mathbf{1656 - 2070}] \text{ m}^2$$

### Number of modules per string

From max and min ambient temperatures, get max and min cell temperatures:

$$T_{\text{cell max}} (^{\circ}\text{C}) = T_{\text{a max}} + 25$$

$$T_{\text{cell max}} (^{\circ}\text{C}) = 45 + 25 = \mathbf{70^{\circ}\text{C}}$$

$$T_{\text{cell min}} (^{\circ}\text{C}) = T_{\text{a min}} + 25$$

$$T_{\text{cell min}} (^{\circ}\text{C}) = 5 + 25 = \mathbf{30^{\circ}\text{C}}$$

Correction of PV module MPP voltage to get  $V_{mpp \max}$  and  $V_{mpp \min}$

For the PV module:

$$V_{mpp \min} = V_{mpp} + [(T_{cell \max} - 25) * T_c * V_{oc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$V_{mpp \min} = 41.75 + [(70 - 25) * -0.304\%/^{\circ}C * 49.54] = \mathbf{35 \text{ V}}$$

$$V_{mpp \max} = V_{mpp} + [(T_{cell \min} - 25) * T_c * V_{oc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$V_{mpp \max} = 41.75 + [(30 - 25) * -0.304\%/^{\circ}C * 49.54] = \mathbf{40.95 \text{ V}}$$

$$\frac{\text{Inverter } V_{mpp \min}}{\text{PV module } V_{mpp \min}} < \text{No. of Modules per String} < \frac{\text{Inverter } V_{mpp \max}}{\text{PV module } V_{mpp \max}}$$

$$\frac{600 \text{ V}}{35 \text{ V}} = 17.14 < \text{No. of Modules per String} < 35.41 = \frac{1450 \text{ V}}{40.95 \text{ V}}$$

$$\mathbf{17 \leq \text{No. of Modules per String} \leq 35}$$

Correction of PV module open circuit voltage to get  $V_{OC \max}$

For the PV module:

$$V_{oc \max} = V_{oc} + [(T_{cell \min} - 25) * T_c * V_{oc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$V_{oc \max} = 49.54 + [(30 - 25) * -0.304\%/^{\circ}C * 49.54] = \mathbf{48.75 \text{ V}}$$

$$\text{No. of Modules per String} < \frac{\text{Inverter } V \max}{\text{PV module } V_{oc \max}}$$

$$\text{No. of Modules per String} < 29.13 = \frac{1420 \text{ V}}{48.75 \text{ V}}$$

$$\text{No. of Modules per String} \leq 29$$

$$\text{No. of Modules per String} = 24$$

## Number of strings

Correction of PV module short circuit current to get  $I_{sc \max}$ .

For the PV module:

$$I_{sc \max} = I_{sc} + [(T_{cell \max} - 25) * T_c * I_{sc}] \quad \text{When } T_c (\%/^{\circ}C)$$

$$I_{sc \max} = 13.89 + [(70 - 25) * 0.050\%/^{\circ}C * 13.89] = \mathbf{14.2 \text{ Amp}}$$

$$\text{No. of Strings per MPPT} < \frac{\text{Inverter } I \max}{\text{PV module } I_{sc \max}}$$

$$\text{No. of Strings per MPPT} < 1.55 = \frac{22 \text{ Amp}}{14.2 \text{ Amp}}$$

$$\text{No. of Strings per MPPT} \leq 1$$

$$\text{Inverter total strings} = 1 \frac{\text{string}}{\text{MPPT}} * \text{no. of MPPTs} = 1 * 4 = 4 \text{ strings}$$

.....

## PV System Summary

PV System Power = **207 kW**

No. of Modules = 207 kW / 540 W = **384** Module

1. No. of Modules in series = **24** Module

2. No. of strings in parallel = **16** String

System Voltage = 24 \* 40.95 = **983 V**

System Current = 4 \* 12.94 = **52 Amp**

No. of Inverters = 207 kW / 60 kW = **4** Inverters

Total Power of Inverters = **240 kW AC**

$\eta_{PV} = 20.9 \%$

$\eta_{Inv} = 99 \%$

$\eta_{Cables} = 98 \%$

Utilized Area = 2514 **m<sup>2</sup>**

Available area percentage = 80 %

PV System Area = 2011 **m<sup>2</sup>**

kW of PV modules cost = **\$1000 / kW**

PV system cost = **\$207,000**

Annual Energy Production = **341,100 kWh**

Annual Saving = Energy \* Tariff = 341,100 \* 1.15 (kWh/LE) = **392,265 LE**

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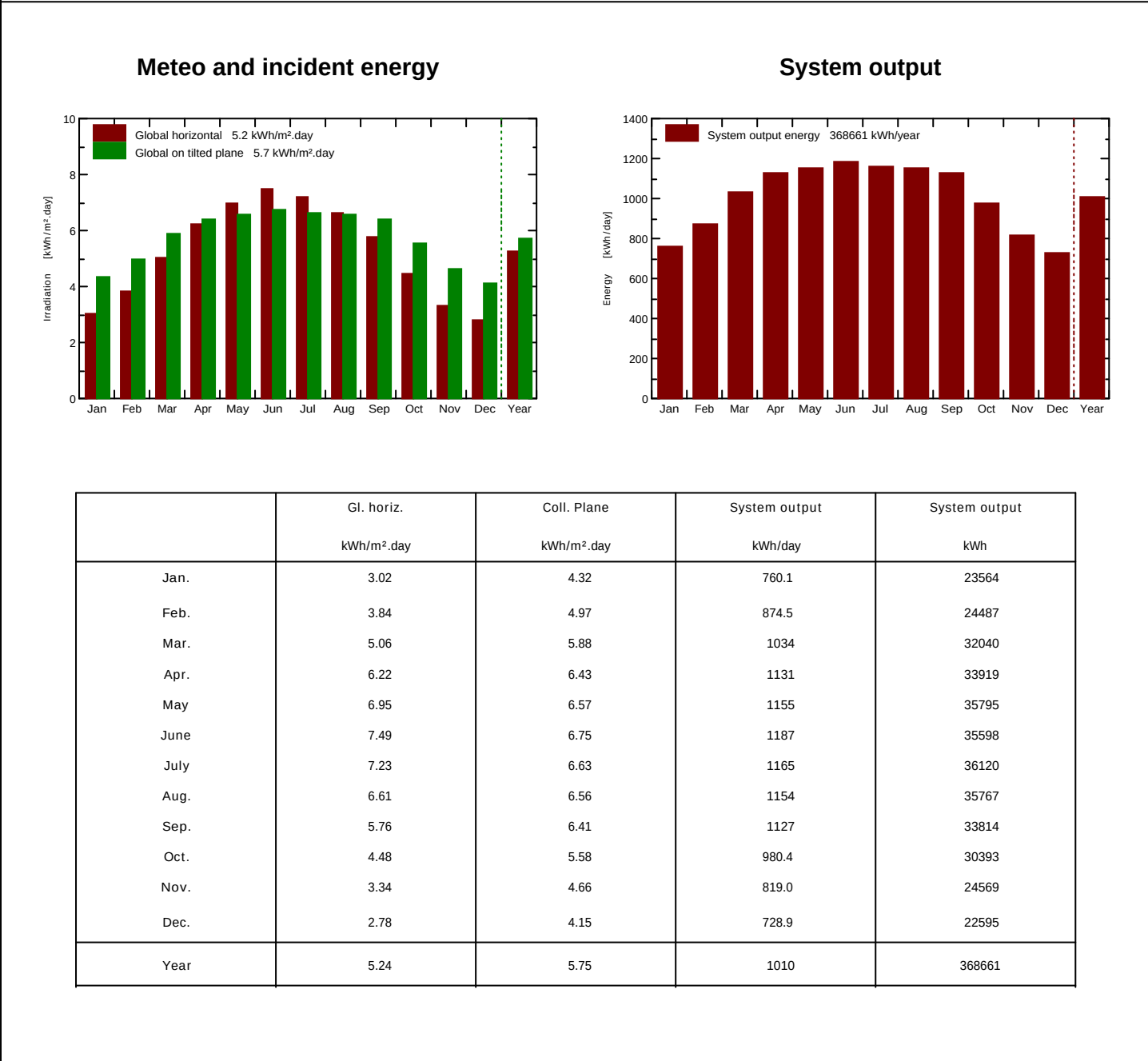
## PVsyst Report

# Grid system presizing

|                                    |                           |           |              |
|------------------------------------|---------------------------|-----------|--------------|
| <b>Geographical Site</b>           | <b>Cairo</b>              | Country   | <b>Egypt</b> |
| <b>Situation</b>                   | Latitude 30.08° N         | Longitude | 31.29° E     |
| Time defined as                    | Legal Time Time zone UT+2 | Altitude  | 0 m          |
| <b>Collector Plane Orientation</b> | Tilt 28°                  | Azimuth   | 5°           |

|                                            |                       |
|--------------------------------------------|-----------------------|
| <b>PV-field installation main features</b> |                       |
| Module type                                | Standard              |
| Technology                                 | Monocrystalline cells |
| Mounting method                            | Flat roof             |
| Back ventilation properties                | Free standing         |

|                                                         |            |            |                             |
|---------------------------------------------------------|------------|------------|-----------------------------|
| <b>System characteristics and pre-sizing evaluation</b> |            |            |                             |
| PV-field nominal power (STC)                            | Pnom       | 207 kWp    |                             |
| Collector area                                          | Acoll      | 1294 m²    |                             |
| Annual energy yield                                     | Eyear      | 369 MWh    | Specific yield 1781 kWh/kWp |
| Economic gross evaluation                               | Investment | 468306 EUR | Energy price 0.08 EUR/kWh   |



|                                              |  |                                                                                 |                                  |                     |                  |
|----------------------------------------------|--|---------------------------------------------------------------------------------|----------------------------------|---------------------|------------------|
| PVSYST V6.81                                 |  |                                                                                 |                                  | 04/05/23            | Page 1/5         |
| Grid-Connected System: Simulation parameters |  |                                                                                 |                                  |                     |                  |
| Project :                                    |  | Tawheed GP 1                                                                    |                                  |                     |                  |
| Geographical Site                            |  | Faculty of Law, Ain Shams University                                            |                                  | Country             | Egypt            |
| Situation                                    |  | Latitude                                                                        | 30.08° N                         | Longitude           | 31.29° E         |
| Time defined as                              |  | Legal Time                                                                      | Time zone UT+2                   | Altitude            | 0 m              |
|                                              |  | Albedo                                                                          | 0.20                             |                     |                  |
| Meteo data:                                  |  | Faculty of Law, Ain Shams University      Meteonorm 7.2 (1981-2009) - Synthetic |                                  |                     |                  |
| Simulation variant :                         |  | New simulation variant                                                          |                                  |                     |                  |
|                                              |  | Simulation date                                                                 | 04/05/23 21h49                   |                     |                  |
| Simulation parameters                        |  | System type                                                                     | No 3D scene defined, no shadings |                     |                  |
| Collector Plane Orientation                  |  | Tilt                                                                            | 28°                              | Azimuth             | 5°               |
| Models used                                  |  | Transposition                                                                   | Perez                            | Diffuse             | Perez, Meteonorm |
| Horizon                                      |  | Free Horizon                                                                    |                                  |                     |                  |
| Near Shadings                                |  | No Shadings                                                                     |                                  |                     |                  |
| User's needs :                               |  | Unlimited load (grid)                                                           |                                  |                     |                  |
| Power factor                                 |  | Cos(phi)                                                                        | 0.950 lagging                    | Phi                 | -18.2°           |
|                                              |  |                                                                                 |                                  |                     |                  |
| PV Array Characteristics                     |  |                                                                                 |                                  |                     |                  |
| PV module                                    |  | Si-mono                                                                         | Model                            | STP550S-C72/Vmh     |                  |
| Custom parameters definition                 |  | Manufacturer                                                                    | Suntech                          |                     |                  |
| Number of PV modules                         |  | In series                                                                       | 24 modules                       | In parallel         | 16 strings       |
| Total number of PV modules                   |  | Nb. modules                                                                     | 384                              | Unit Nom. Power     | 540 Wp           |
| Array global power                           |  | Nominal (STC)                                                                   | 207 kWp                          | At operating cond.  | 185 kWp (50°C)   |
| Array operating characteristics (50°C)       |  | U mpp                                                                           | 891 V                            | I mpp               | 207 A            |
| Total area                                   |  | Module area                                                                     | 992 m²                           | Cell area           | 671 m²           |
| Inverter                                     |  | Model                                                                           | SUN2000-60KTL-HV-D1-001          |                     |                  |
| Original PVsyst database                     |  | Manufacturer                                                                    | Huawei Technologies              |                     |                  |
| Characteristics                              |  | Operating Voltage                                                               | 600-1450 V                       | Unit Nom. Power     | 60.0 kWac        |
|                                              |  |                                                                                 |                                  | Max. power (=>30°C) | 66.0 kWac        |
| Inverter pack                                |  | Nb. of inverters                                                                | 4 units                          | Total Power         | 240 kWac         |
|                                              |  |                                                                                 |                                  | Pnom ratio          | 0.86             |
|                                              |  |                                                                                 |                                  |                     |                  |
| PV Array loss factors                        |  |                                                                                 |                                  |                     |                  |
| Array Soiling Losses                         |  |                                                                                 | Loss Fraction                    | 3.0 %               |                  |
| Thermal Loss factor                          |  | Uc (const)                                                                      | 29.0 W/m²K                       | Uv (wind)           | 0.0 W/m²K / m/s  |
| Wiring Ohmic Loss                            |  | Global array res.                                                               | 72 mOhm                          | Loss Fraction       | 1.5 % at STC     |
| LID - Light Induced Degradation              |  |                                                                                 |                                  | Loss Fraction       | 1.5 %            |
| Module Quality Loss                          |  |                                                                                 |                                  | Loss Fraction       | -1.3 %           |
| Module Mismatch Losses                       |  |                                                                                 |                                  | Loss Fraction       | 1.0 % at MPP     |
| Strings Mismatch loss                        |  |                                                                                 |                                  | Loss Fraction       | 0.10 %           |
| Incidence effect, ASHRAE parametrization     |  | IAM =                                                                           | 1 - bo (1/cos i - 1)             | bo Param.           | 0.05             |
| System loss factors                          |  |                                                                                 |                                  |                     |                  |
| Wiring Ohmic Loss                            |  | Wires: 3x50.0 mm²                                                               | 42 m                             | Loss Fraction       | 0.5 % at STC     |
| Unavailability of the system                 |  | 1.0 days, 3 periods                                                             |                                  | Time fraction       | 0.3 %            |

## Grid-Connected System: Main results

**Project :** Tawheed GP 1  
**Simulation variant :** New simulation variant

### Main system parameters

PV Field Orientation

PV modules

PV Array

Inverter

Inverter pack

User's needs

System type

**No 3D scene defined, no shadings**

tilt

Model

Nb. of modules

Model

Nb. of units

Unlimited load (grid)

28°

azimuth 5°

STP550S-C72/Vmh

384

SUN2000-60KTL-HV-D1-001

4.0

Pnom

Pnom total

Pnom

Pnom total

Cos(Phi)

540 Wp

**207 kWp**

60.0 kW ac

**240 kW ac**

0.950 lagging

### Main simulation results

System Production

**Produced Energy**

Apparent energy

**341.1 MWh/year**

359.0 MVAh

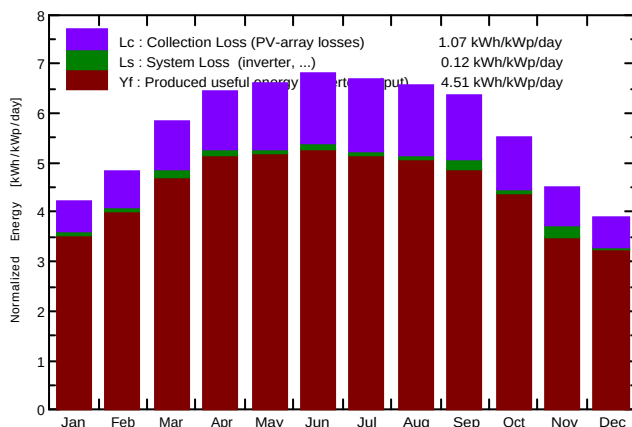
Specific prod.

Perf. Ratio PR

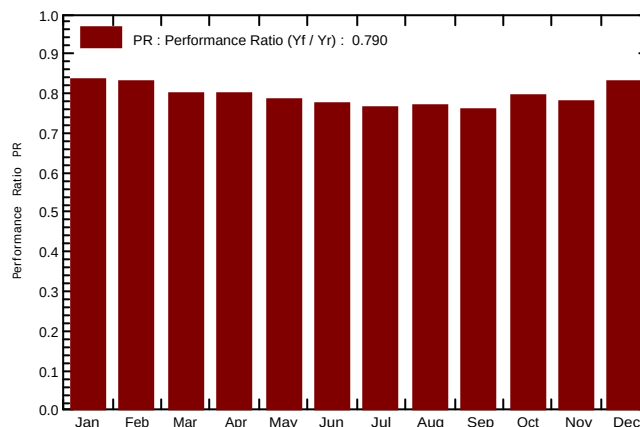
1645 kWh/kWp/year

79.04 %

Normalized productions (per installed kWp): Nominal power 207 kWp



Performance Ratio PR



### New simulation variant Balances and main results

|           | GlobHor<br>kWh/m <sup>2</sup> | DiffHor<br>kWh/m <sup>2</sup> | T_Amb<br>°C | GlobInc<br>kWh/m <sup>2</sup> | GlobEff<br>kWh/m <sup>2</sup> | EArray<br>MWh | E_Grid<br>MWh | PR    |
|-----------|-------------------------------|-------------------------------|-------------|-------------------------------|-------------------------------|---------------|---------------|-------|
| January   | 93.5                          | 39.44                         | 14.53       | 131.3                         | 124.0                         | 23.18         | 22.76         | 0.836 |
| February  | 107.5                         | 50.65                         | 15.55       | 135.5                         | 128.0                         | 23.85         | 23.42         | 0.833 |
| March     | 157.1                         | 65.52                         | 18.66       | 181.4                         | 171.4                         | 31.40         | 30.21         | 0.803 |
| April     | 186.8                         | 77.40                         | 21.60       | 193.4                         | 182.1                         | 32.77         | 32.15         | 0.802 |
| May       | 215.7                         | 84.51                         | 25.51       | 204.9                         | 192.6                         | 33.95         | 33.28         | 0.783 |
| June      | 225.1                         | 76.87                         | 27.95       | 204.7                         | 192.2                         | 33.48         | 32.81         | 0.773 |
| July      | 224.3                         | 80.45                         | 29.57       | 207.8                         | 195.1                         | 33.72         | 33.04         | 0.767 |
| August    | 205.2                         | 74.43                         | 29.36       | 204.0                         | 192.0                         | 33.23         | 32.56         | 0.770 |
| September | 172.8                         | 60.58                         | 27.36       | 191.4                         | 180.7                         | 31.60         | 30.20         | 0.761 |
| October   | 139.0                         | 57.41                         | 24.44       | 170.4                         | 161.2                         | 28.61         | 28.06         | 0.794 |
| November  | 100.1                         | 44.61                         | 19.75       | 135.1                         | 127.7                         | 23.28         | 21.81         | 0.778 |
| December  | 86.1                          | 42.09                         | 16.22       | 120.9                         | 114.1                         | 21.16         | 20.78         | 0.829 |
| Year      | 1913.2                        | 753.96                        | 22.58       | 2081.0                        | 1961.2                        | 350.24        | 341.08        | 0.790 |

Legends:

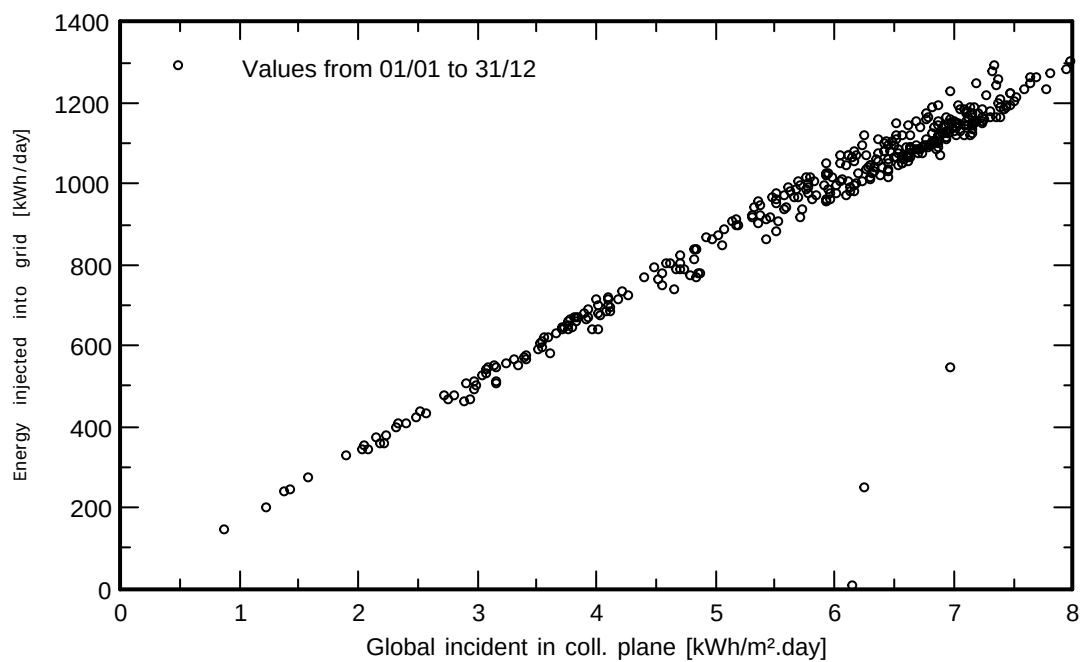
|         |                                |         |                                              |
|---------|--------------------------------|---------|----------------------------------------------|
| GlobHor | Horizontal global irradiation  | GlobEff | Effective Global, corr. for IAM and shadings |
| DiffHor | Horizontal diffuse irradiation | EArray  | Effective energy at the output of the array  |
| T_Amb   | Ambient Temperature            | E_Grid  | Energy injected into grid                    |
| GlobInc | Global incident in coll. plane | PR      | Performance Ratio                            |

Grid-Connected System: Special graphs

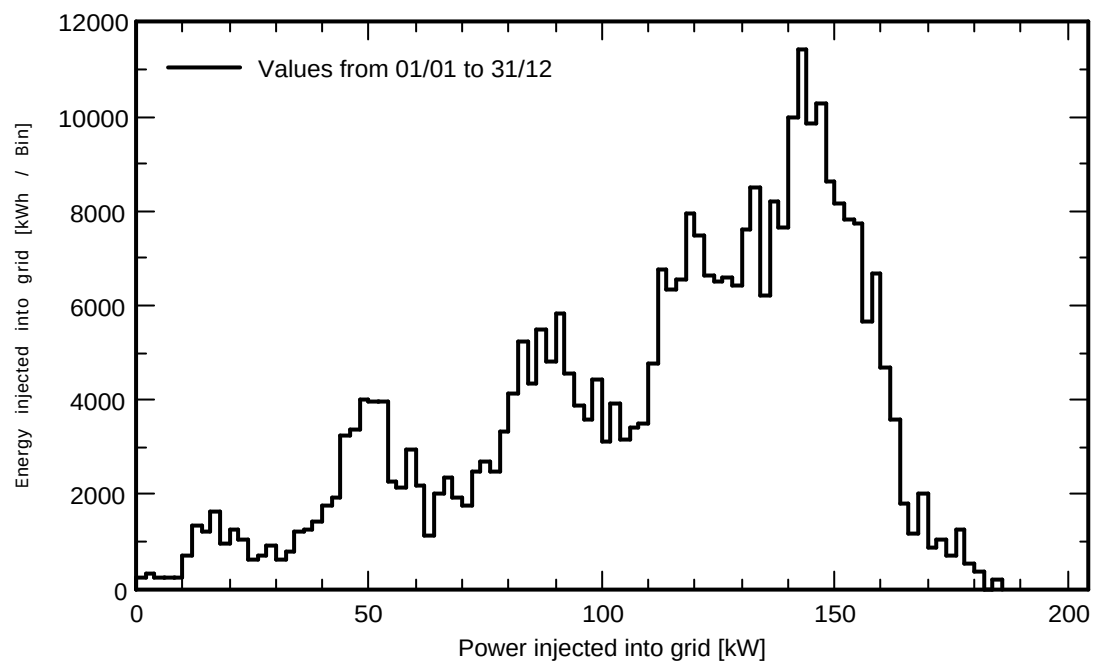
Project : Tawheed GP 1  
Simulation variant : New simulation variant

|                               |  |                       |                                         |                             |
|-------------------------------|--|-----------------------|-----------------------------------------|-----------------------------|
| <b>Main system parameters</b> |  | System type           | <b>No 3D scene defined, no shadings</b> |                             |
| PV Field Orientation          |  | tilt                  | 28°                                     | azimuth 5°                  |
| PV modules                    |  | Model                 | STP550S-C72/Vmh                         | Pnom 540 Wp                 |
| PV Array                      |  | Nb. of modules        | 384                                     | Pnom total <b>207 kWp</b>   |
| Inverter                      |  | Model                 | SUN2000-60KTL-HV-D1-001                 | Pnom 60.0 kW ac             |
| Inverter pack                 |  | Nb. of units          | 4.0                                     | Pnom total <b>240 kW ac</b> |
| User's needs                  |  | Unlimited load (grid) |                                         | Cos(Phi) 0.950 lagging      |

Daily Input/Output diagram



System Output Power Distribution



## Grid-Connected System: Loss diagram

**Project :** Tawheed GP 1

**Simulation variant :** New simulation variant

### Main system parameters

PV Field Orientation

PV modules

PV Array

Inverter

Inverter pack

User's needs

System type

tilt

**No 3D scene defined, no shadings**

28°

azimuth 5°

Model

STP550S-C72/Vmh

Pnom

540 Wp

Nb. of modules

384

Pnom total

**207 kWp**

Model

SUN2000-60KTL-HV-D1-001

Pnom

60.0 kW ac

Nb. of units

4.0

Pnom total

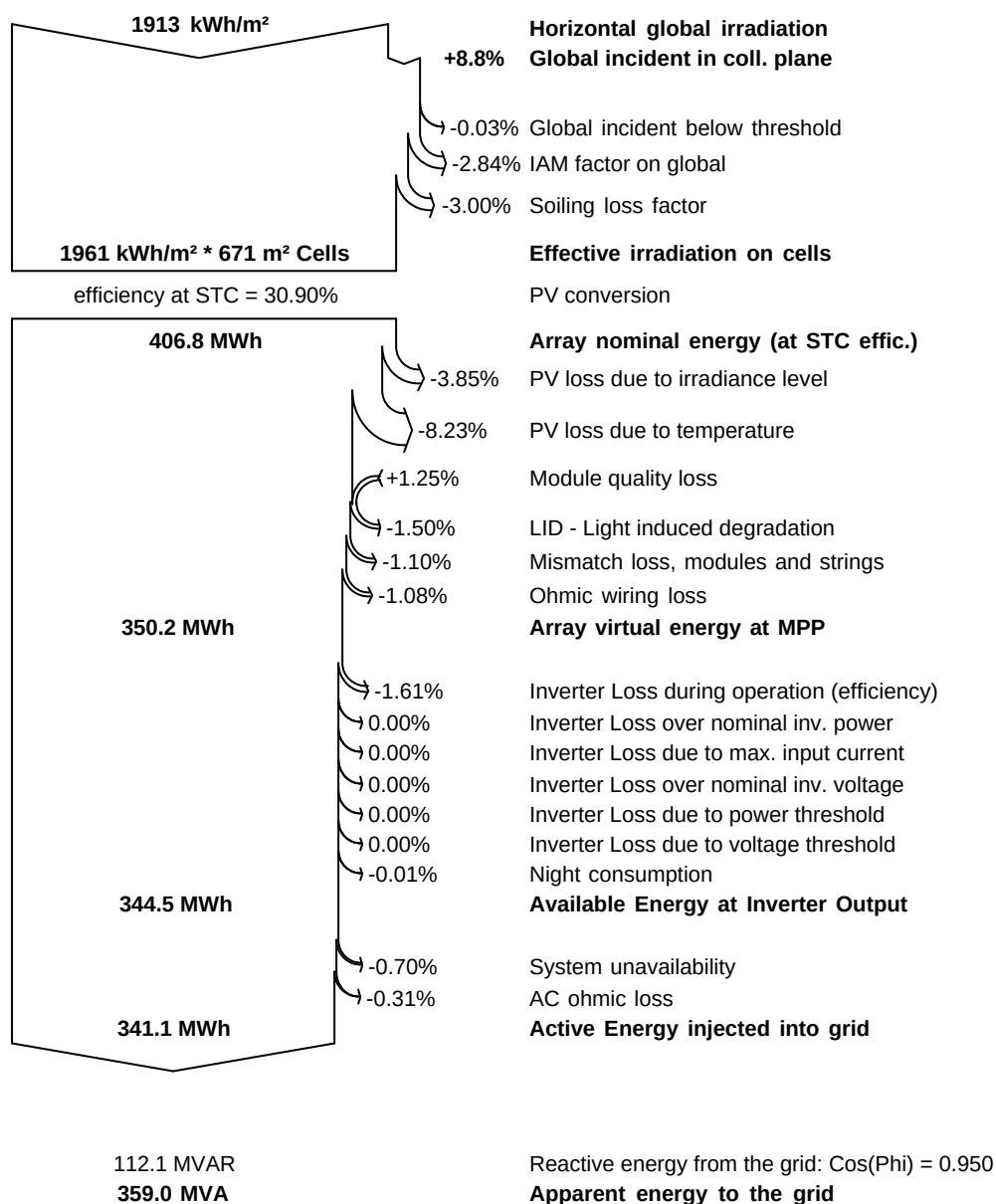
**240 kW ac**

Unlimited load (grid)

Cos(Phi)

0.950 lagging

### Loss diagram over the whole year



## Grid-Connected System: P50 - P90 evaluation

**Project :** Tawheed GP 1

**Simulation variant :** New simulation variant

|                               |                       |                                         |            |                  |
|-------------------------------|-----------------------|-----------------------------------------|------------|------------------|
| <b>Main system parameters</b> | System type           | <b>No 3D scene defined, no shadings</b> |            |                  |
| PV Field Orientation          | tilt                  | 28°                                     | azimuth    | 5°               |
| PV modules                    | Model                 | STP550S-C72/Vmh                         | Pnom       | 540 Wp           |
| PV Array                      | Nb. of modules        | 384                                     | Pnom total | <b>207 kWp</b>   |
| Inverter                      | Model                 | SUN2000-60KTL-HV-D1-001                 | Pnom       | 60.0 kW ac       |
| Inverter pack                 | Nb. of units          | 4.0                                     | Pnom total | <b>240 kW ac</b> |
| User's needs                  | Unlimited load (grid) |                                         | Cos(Phi)   | 0.950 lagging    |

### Evaluation of the Production probability forecast

The probability distribution of the system production forecast for different years is mainly dependent on the meteo data used for the simulation, and depends on the following choices:

|                          |                           |             |           |
|--------------------------|---------------------------|-------------|-----------|
| Meteo data source        | Meteonorm 7.2 (1981-2009) |             |           |
| Meteo data               | Kind                      | Not defined | Year 1995 |
| Specified Deviation      | Year deviation from aver. | 3 %         |           |
| Year-to-year variability | Variance                  | 0.5 %       |           |

The probability distribution variance is also depending on some system parameters uncertainties

|                                     |                                    |       |                 |
|-------------------------------------|------------------------------------|-------|-----------------|
| Specified Deviation                 | PV module modelling/parameters     | 1.0 % |                 |
|                                     | Inverter efficiency uncertainty    | 0.5 % |                 |
|                                     | Soiling and mismatch uncertainties | 1.0 % |                 |
|                                     | Degradation uncertainty            | 1.0 % |                 |
| Global variability (meteo + system) | Variance                           | 1.9 % | (quadratic sum) |

|                               |                    |                  |
|-------------------------------|--------------------|------------------|
| Annual production probability | <b>Variability</b> | <b>6.4 MWh</b>   |
|                               | <b>P50</b>         | <b>341.1 MWh</b> |
|                               | <b>P90</b>         | <b>332.9 MWh</b> |
|                               | <b>P95</b>         | <b>330.6 MWh</b> |

**Probability distribution**

